GSSSB · STAFF NURSE EXAM

Community Health Nursing (CHN)

Complete Colourful Notes · 3D Diagrams · Mnemonics · Exam Points



17 Chapters | 3D Coloured Diagrams | Epidemiology · Biostatistics | National Health Programmes | Health care delivery | Exam Points (PYQ style)

Hinglish explanation + English exam terms

Sister Asha
Aapki Nursing teacher

Index – Kya padhenge?

1 **Concepts of Health, Disease & Community Health Nursing**
Definitions of health, dimensions, determinants, spectrum, concepts of disease causation, natural history, levels of prevention, community, CHN principles & roles

2 **Health Care Delivery System in India**
Health committees, Alma-Ata & primary health care, health system levels, population norms, Ayushman Bharat (AAM & PM-JAY), NHM, NHP 2017, health team, IPHS

3 **Epidemiology**
Definitions, aims, epidemiological triad, measurements (rates, ratios, proportions), incidence & prevalence, study designs, screening (sensitivity, specificity), epidemic investigation

4 **Biostatistics, Demography & Vital Statistics**
Data types, presentation, measures of central tendency & dispersion, normal curve, sampling, tests, census, registration, vital rates, demographic cycle, India's health indicators

5 **Communicable Disease Epidemiology & Control**
Sources & reservoirs, carriers, modes of transmission, incubation, isolation, quarantine, disinfection, surveillance (IDSP/ IHIP), notifiable diseases, outbreak control

6 **National Programmes – Communicable Diseases**
NTEP (TB), NLEP (leprosy), NCVBDC (malaria, dengue, filaria, kala-azar, JE, chikungunya), NACP (HIV), viral hepatitis, polio, rabies, COVID-19, AMR, cholera & typhoid

7 **National Programmes – NCDs & Others**
NP-NCD (NPCDCS), cancer screening, NPCBVI, NMHP & Tele-MANAS, NTCP & COTPA, NPHCE, NPPCD, NIDDCP, fluorosis, oral health, burns, climate change, IDD, important days

8 **RMNCH+A, Family Welfare & Population Control**
RMNCH+A strategy, maternal & child health services in community, schemes (JSY, JSSK, PMSMA, PMMVY), family planning methods & programme, MTP, PCPNDT, adolescent health

9 **Nutrition in Community Health**
Nutrients & RDA, balanced diet, malnutrition burden, nutritional assessment, food adulteration, food fortification, nutrition programmes (ICDS, PM POSHAN, POSHAN Abhiyaan, TPDS), food safety

10 **Environmental Health & Sanitation**
Water – sources, purification, chlorination, well disinfection, water-borne diseases; air pollution; excreta & refuse disposal; biomedical waste rules; housing, noise, radiation; medical entomology; Swachh Bharat

11 **Occupational Health**
Occupational hazards, pneumoconiosis, lead & other poisonings, occupational cancers & dermatoses, prevention, ESI Act, Factories Act, health of workers, ergonomics

12 **Health Education, Communication & IEC/ BCC**
Definition, aims, principles, approaches, contents, methods (individual, group, mass), AV aids, Edgar Dale's cone, communication process & barriers, counselling, models of behaviour change

13 **Home Visit, Family Health Care, Records & Reports**
Family as unit, family health nursing process, community diagnosis & survey, home visit principles & steps, bag technique, clinics, records, registers, reports, referral, ANM/ CHO roles

14 **Health of Special Groups – School, Urban, Tribal, Elderly, Disabled & Others**
School health services, Ayushman Bharat School Health, urban health & slums, tribal health, geriatric care, rehabilitation & CBR, mental health in community, prison/ industrial health, migrants, women empowerment

15 **Disaster Management & Emergency Services**
Types of disasters, disaster cycle, DM Act 2005, NDMA/ NDRF/ SDMA, triage, mass casualty management, nurse's role, epidemics after disasters, first aid, 108 EMRI, emergency preparedness

16 **Health Planning, Management, International Agencies & Health Insurance**
Planning cycle, management techniques, five-year plans & NITI Aayog, supervision, international health agencies (WHO, UNICEF, ILO etc.), voluntary agencies, health insurance & financing, Gujarat health schemes

17 **Quick Revision – Master Tables & 150 One-Liners**
All community health numbers, years, committees, programmes, days, formulas & 150 exam one-liners in one place

★ **Key Points** – sabse important facts

Asha Didi ki Tip – samjhane wali baat

! Nursing Alert – galti mat karna

✎ **Exam Point** – PYQ style, baar baar aata hai

♫ **Mnemonic** – yaad karne ki trick

D Definition – exact definition

1 Concepts of Health, Disease & Community Health Nursing

1.1 Health

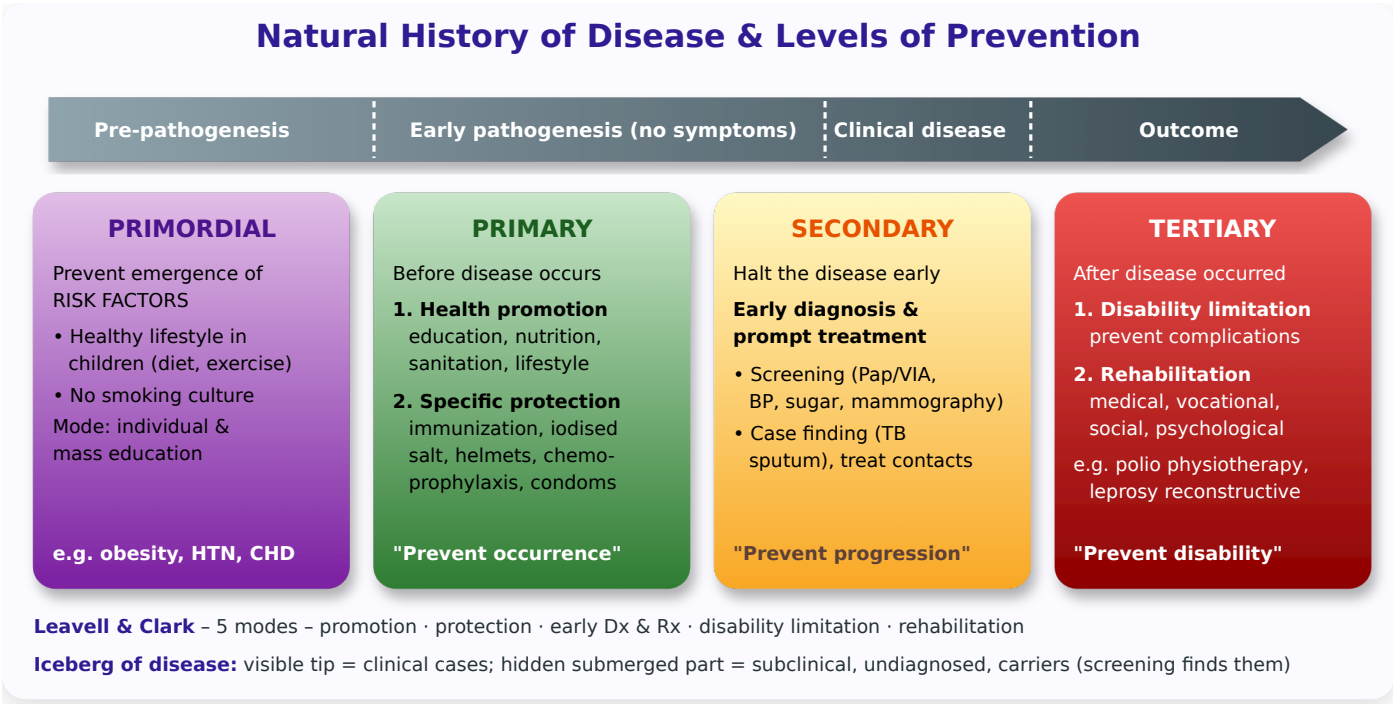
D Health (WHO, 1948)

"Health is a state of **complete physical, mental and social well-being** and not merely the absence of disease or infirmity." **Spiritual dimension** suggested later (1980s). **Operational definition** – absence of obvious disease + functioning normally.

- **Dimensions of health** – **physical, mental, social, spiritual**, emotional, vocational, (environmental, philosophical, cultural, socioeconomic, educational, nutritional, curative, preventive).
- **Positive health** – perfect functioning of body & mind (ideal, not achievable fully). **Health continuum / spectrum** – positive health → better health → freedom from sickness → unrecognised sickness → mild → severe sickness → death.
- **Concepts of health** – **biomedical** (absence of disease – germ theory), **ecological** (man–environment equilibrium), **psychosocial**, **holistic** (all sectors – most accepted).
- **Determinants of health** – **biological (heredity)**, **behavioural & socio-cultural (lifestyle)**, **environment**, **socioeconomic** (income, education, occupation, political system), **health services**, ageing, gender, equity, information & communication, aging of population, other sectors (agriculture, food).
- **Indicators of health** – mortality (**IMR – most sensitive**), morbidity, disability (**DALY** – disability-adjusted life year = YLL + YLD; **Sullivan's index** – expectation of life free of disability; **QALY**), nutritional status, health care delivery (doctor–population ratio), utilisation (immunization coverage), social & mental health, environmental, socio-economic, health policy, quality of life (**PQLI – Physical Quality of Life Index = IMR + life expectancy at age 1 + literacy**; 0–100), **HDI (Human Development Index – life expectancy at birth, education – mean & expected years of schooling, GNI per capita)** – by UNDP.
- **Responsibility for health** – individual (self-care), community, state, international.

1.2 Disease & Its Causation

- **Disease** – a condition in which body health is impaired; **illness** – subjective (patient's feeling); **sickness** – social role (state of social dysfunction).
- **Theories of disease causation**: **supernatural** → **theory of humours (Hippocrates)** → **miasmatic theory** (bad air) → **germ theory (Louis Pasteur, Robert Koch)** – single cause → **epidemiological triad (agent–host–environment)** → **multifactorial causation** (Pettenkofer) → **web of causation (MacMahon & Pugh) – for chronic diseases**.
- **Koch's postulates** – agent present in every case, isolated in pure culture, reproduces disease in animals, re-isolated.
- **Iceberg phenomenon of disease** – **tip = clinical cases (seen)**; submerged portion = **sub-clinical, carriers, undiagnosed** (larger); e.g. **hypertension, diabetes, anaemia, malnutrition, polio** (large hidden). **Diseases without iceberg** – rabies, tetanus, measles (mostly clinical).



Natural history of disease & the four levels of prevention

1.3 Natural History of Disease

- **Pre-pathogenesis phase** – disease agent not yet entered man; interaction of agent, host, environment (**man in the presence of the disease** – potential).
- **Pathogenesis phase** – entry of agent → **incubation period** → early pathogenesis (subclinical) → clinical disease → **outcome (recovery, disability, carrier state, death)**.

1.4 Levels of Prevention (Leavell & Clark)

Level	Phase	Modes of intervention	Examples
Primordial	Before risk factors appear	Health education (lifestyle)	Preventing obesity, smoking habit in children
Primary	Pre-pathogenesis	**Health promotion** (education, nutrition, environment, lifestyle) & **specific protection**	**Immunization, chemoprophylaxis, iodised salt, fluoridation, helmet, protective clothing, condoms**
Secondary	Early pathogenesis	**Early diagnosis & treatment**	**Screening (Pap smear, BP, sugar), case finding, treatment of TB/ leprosy patients**
Tertiary	Late pathogenesis	**Disability limitation & rehabilitation**	Physiotherapy in polio, reconstructive surgery in leprosy, CBR

- **Rehabilitation types** – **medical** (restoration of function), **vocational** (earning capacity), **social** (family/ social relationships), **psychological** (personal dignity, confidence).
- **Disability chain** – **disease → impairment → disability → handicap** (WHO ICIDH); impairment = loss of structure/ function (organ level); disability = restriction of activity (person level); handicap = social disadvantage. **ICF (2001)** – functioning, disability & health.
- **Primary prevention most effective & economical**; **secondary prevention = main approach for chronic diseases without primary prevention**.

1.5 Community & Community Health Nursing

- **Community** – a group of people living in a **common geographical area**, sharing common values, interests, culture, needs (WHO). Characteristics – common area, common interests, sense of belonging, social structure.
- **Community health** – health status of members of a community and actions to improve it.
- **Community health nursing (CHN)** – synthesis of **nursing practice and public health practice** applied to promoting & preserving health of populations; **population-focused, comprehensive, continuous, family as unit of service**.
- **Public health (Winslow, 1920)** – "the science & art of **preventing disease, prolonging life and promoting health & efficiency** through organised community effort".
- **Father of public health** – **Cholera John Snow (father of modern epidemiology)**; Father of public health nursing – **Lillian Wald** (coined term "public health nurse", Henry Street Settlement, New York 1893); **William Rathbone – district nursing (Liverpool, 1859)**; **Florence Nightingale – "health nursing"**.
- **Principles of CHN** – based on **needs of community**, **family as unit**, **community participation**, health education as integral part, **continuity of care**, utilise existing resources, **intersectoral coordination**, periodic evaluation, records & reports, work within **national health policy**, respect for cultural values, **preventive & promotive focus**.

Roles of Community Health Nurse

- **Care provider** (clinic, home), **health educator**, **counsellor**, **advocate**, **case manager/ coordinator**, **collaborator**, **leader/ manager/ supervisor** (of ANMs, ASHAs), **researcher**, **change agent**, **epidemiologist/ case finder** (surveillance), **referral agent**, **trainer** (of health workers, dais), **record keeper**.
- **Qualities** – good communicator, culturally sensitive, observant, resourceful, empathetic.

1.6 Health Promotion & Concepts

- **Ottawa Charter (1986)** – first International Conference on Health Promotion; **5 action areas** – build healthy public policy, create supportive environments, strengthen community action, develop personal skills, reorient health services. **Strategies – advocate, enable, mediate**.
- **Health promotion** – process of enabling people to increase control over & improve their health.
- **Concepts of control** – **eradication** (termination of all transmission by extermination of agent – **smallpox (1980)**, globally), **elimination** (interruption of transmission from a large region – **regional/ national**, e.g. polio from India, yaws), **control** (reduction of incidence to a level where it is no longer a public health problem), **monitoring** & **surveillance**.
- **Smallpox – only human disease eradicated globally (WHO declared 8 May 1980)**; India smallpox-free – **1977** (last case Saiban Bibi – 24 May 1975 – Assam/Karimganj); **Yaws** – India declared free 2006 (first country WHO-certified 2016); **Guinea worm** – India certified free 2000; **Polio** – 2014; **Maternal & neonatal tetanus** – 2015; **Trachoma** – WHO validated India as having eliminated trachoma as a public health problem in **2024**.

PYQ style

- WHO health definition year – **1948**; dimensions – **physical, mental, social (spiritual added later)**.
- Most sensitive indicator of health – **IMR**.
- PQLI includes – **IMR, life expectancy at age 1, literacy**.
- HDI is calculated by – **UNDP**.
- DALY = **YLL + YLD**.
- Web of causation – **chronic diseases (MacMahon & Pugh)**.
- Iceberg phenomenon – **hypertension, diabetes**.
- Immunization is – **primary prevention (specific protection)**.
- Pap smear screening – **secondary prevention**.
- Physiotherapy in polio – **tertiary prevention (rehabilitation)**.
- Primordial prevention – **prevent risk factors**.
- Father of public health nursing – **Lillian Wald**; district nursing – **William Rathbone**.
- Ottawa Charter – **1986 (health promotion)**.
- Only disease eradicated – **smallpox (1980)**.

2 Health Care Delivery System in India

2.1 Health Committees (Most Asked)

Committee	Year	Key recommendation
Bhore Committee (Health Survey & Development Committee, chairman **Sir Joseph Bhore**)	**1946** (appointed 1943)	**Integration of curative & preventive**; **PHC concept**; 3-month training in preventive medicine; **"social physician"**; short-term: **1 PHC per 40,000** (long-term **3 Million Plan** – 75-bed PHC per 10,000–20,000)
Mudaliar Committee (Health Survey & Planning Committee)	**1962**	Strengthen existing PHCs before new; **1 PHC per 40,000**; District hospitals strengthened; **All India Health Service**
Chadah Committee	**1963**	Malaria maintenance phase – **1 basic health worker per 10,000** population (vigilance); family planning duty to health workers
Mukerji Committee	**1965 / 1966**	Separate staff for family planning; delinking malaria from FP
Jungalwalla Committee (Integration)	**1967**	**Integration of health services**, unified cadre, equal pay, no private practice
Kartar Singh Committee (Multipurpose workers)	**1973**	**Multipurpose workers (MPW)** – ANM → **female health worker**; male MPW; **1 PHC per 50,000** with **16 sub-centres (each ~3,000–3,500)** having 1 male + 1 female worker; **1 health supervisor per 4 health workers**
Shrivastava Committee (Group on Medical Education & Support Manpower)	**1975**	**Community health workers/ Village health guides**; referral services complex (**ROME – Reorientation of Medical Education**, 1977)
Bajaj Committee	1986	Health manpower planning

- **Village Health Guide Scheme – 2 Oct 1977**; **ICDS – 1975**; **ROME scheme 1977**.

2.2 Alma-Ata Declaration & Primary Health Care

D **Alma-Ata (Almaty, Kazakhstan; USSR) – 6–12 Sept 1978**

WHO-UNICEF International Conference on Primary Health Care; goal **"Health for All by 2000 AD"**; **Primary Health Care** = essential health care based on practical, scientifically sound & socially acceptable methods, **universally accessible** to individuals & families **through their full participation**, at a **cost the community & country can afford**. **Astana Declaration 2018** – renewed commitment.

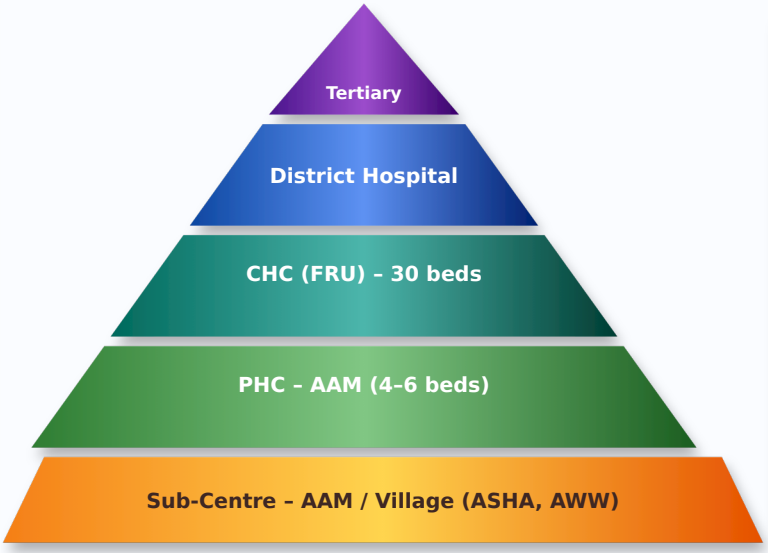
- **Elements of PHC (8 – "ELEMENTS")**: **E**ducation (health education on prevailing problems), **L**ocally endemic disease control, **E**xpanded immunization, **M**aternal & child health incl. family planning, **E**ssential drugs, **N**utrition (food supply & proper nutrition), **T**reatment of common diseases & injuries, **S**afe water & basic sanitation. (+ mental health, dental, handicapped – added later).
- **Principles of PHC (4 + 1)**: **Equitable distribution**, **community participation**, **intersectoral coordination**, **appropriate technology** (+ focus on prevention).

- India accepted – **National Health Policy 1983**.

2.3 Health Care Delivery – Structure

- **Central level** – **Ministry of Health & Family Welfare** (Departments: Health & Family Welfare, Health Research/ ICMR); **DGHS** (Directorate General of Health Services – technical adviser); **Central Council of Health & Family Welfare** (1952, chairman – Union Health Minister).
- **State level** – State Ministry, Directorate of Health Services; **health is a State subject** (Constitution – **7th Schedule, State List**; **Article 47 – duty of state to raise nutrition & public health**).
- **District level** – **Chief District Health Officer (CDHO)/ CMO**, district hospital; **district – principal unit of administration**.
- **Block/ Taluka level** – **CHC**, PHCs; **Taluka Health Officer (THO)/ BMO**.
- **Village level** – **sub-centre (AAM)**, ASHA, AWW, **VHSNC (Village Health, Sanitation & Nutrition Committee)**, **Gram Panchayat**; **Panchayati Raj (3-tier – 73rd Amendment 1992)** – Gram Panchayat, Panchayat Samiti (block), Zila Parishad (district).

Rural Health Care Delivery in India – Population Norms



Population covered (Plain / Hilly-tribal)

Sub-Centre	5,000 / 3,000
ANM + MPW (M) + CHO (at AAM/ HWC)	
PHC	30,000 / 20,000
Medical officer · 6 SCs · referral to CHC	
CHC	1,20,000 / 80,000
4 PHCs · specialists (surgeon, physician, OBG, paediatrician) · FRU	
ASHA	1 per 1,000
Woman 25–45 yrs, literate (8th–10th), resident	
AAM / HWC	Ayushman Bharat (2018)
SC & PHC upgraded → CPHC 12 service packages	
Renamed Ayushman Arogya Mandir (2023)	
Village Health Sanitation & Nutrition Committee	

Levels of care: Primary (SC, PHC) → Secondary (CHC, sub-district & district hospitals) → Tertiary (medical colleges, AIIMS)

Bhore Committee (1946) – recommended PHC concept; **Alma-Ata (1978)** – Primary Health Care, "Health for All"

Rural health infrastructure pyramid & population norms (IPHS)

Facility (IPHS 2022)	Plain area	Hilly/ tribal/ difficult	Key points
Sub-Centre (SC) – AAM	**5,000**	**3,000**	**Most peripheral contact point**; 1 ANM (FHW) + 1 MPW (male) + **CHO (Community Health Officer)** at AAM; **6 SCs under 1 PHC**
PHC – AAM	**30,000**	**20,000**	**Medical Officer** + 14 staff; 4–6 beds; **1st contact with doctor**; referral unit for 6 SCs
CHC (FRU)	**1,20,000**	**80,000**	**30 beds**, **4 specialists (surgeon, physician, OBG, paediatrician)** + anaesthetist; **referral for 4 PHCs**; **First Referral Unit** (emergency obstetric care, CS, blood storage, newborn care)

Facility (IPHS 2022)	Plain area	Hilly/ tribal/ difficult	Key points
Sub-district/ district hospital	–	–	Secondary care
Medical colleges, AIIMS	–	–	Tertiary care

- **Urban** – **Urban PHC (UPHC) per 50,000** (NUHM); **UCHC per 2.5–5 lakh**; **Urban AAM (per 15,000–20,000)**; **MAS (Mahila Arogya Samiti)** in slums; **ASHA per 1,000–2,500** in urban slums.
- **Ayushman Bharat Digital Mission (ABDM)** – **ABHA (Ayushman Bharat Health Account) – 14-digit health ID** (Sept 2021).

2.4 Ayushman Bharat (2018)

Pillar	Details
Health & Wellness Centres (HWC) → Ayushman Arogya Mandir (AAM) (renamed **Nov 2023**)	Launched **14 April 2018 – Jangla, Bijapur (Chhattisgarh)**; SCs & PHCs upgraded to deliver **Comprehensive Primary Health Care (CPHC)** – **12 service packages**; **CHO (mid-level health provider – B.Sc. nursing/ GNM/ BAMS with 6-month Certificate in CPHC)**; free essential drugs & diagnostics; **teleconsultation (eSanjeevani)**; wellness activities (yoga)
PM-JAY (Pradhan Mantri Jan Arogya Yojana)	Launched **23 Sept 2018 (Ranchi, Jharkhand)**; **₹5 lakh per family per year** for secondary & tertiary hospitalisation; **cashless, paperless**; beneficiaries per **SECC 2011** deprivation criteria; **no cap on family size or age**; **Ayushman card**; **extended to all senior citizens ≥ 70 yrs irrespective of income (Oct 2024 – Ayushman Vay Vandana card)**; **National Health Authority (NHA)** implements

- **CPHC – 12 service packages**: (1) **care in pregnancy & childbirth**, (2) **neonatal & infant care**, (3) **childhood & adolescent care**, (4) **family planning, contraception & reproductive health**, (5) **management of communicable diseases & national health programmes**, (6) **general OPD – acute simple illnesses & minor ailments**, (7) **screening, prevention & management of NCDs**, (8) **mental health**, (9) **ophthalmic (eye) care**, (10) **ENT care**, (11) **oral health**, (12) **elderly & palliative care** (+ emergency medical services incl. burns & trauma – expanded).
- **Universal screening of NCDs (≥ 30 yrs)** at AAM – hypertension, diabetes, **oral, breast & cervical cancer**.
- **Gujarat** – **PMJAY–MA (Mukhyamantri Amrutum) – cover ₹10 lakh per family** (state top-up); MA scheme started **2012** for BPL.

2.5 National Health Mission (NHM)

- **NRHM – launched 12 April 2005** (rural; 18 high-focus states); **NUHM – 1 May 2013**; both under **NHM (2013)**.
- **Key components** – **ASHA** (1 per 1,000 population), **Rogi Kalyan Samiti (RKS)/ Hospital Management Society**, **untied funds** to SCs/ PHCs/ CHCs, **VHSNC**, **IPHS** standards, **JSY**, mobile medical units, **NAMS**, district health action plans, **RMNCH+A** (2013), AYUSH mainstreaming, **National Ambulance Services (108/ 102)**.

- **ASHA (Accredited Social Health Activist)** – **woman, resident of the village, 25–45 yrs (preferably married/ widowed/ divorced), literate – formal education up to 8th class (10th preferred)**, selected by **Gram Sabha/ VHSNC**; **1 per 1,000 population** (tribal/ hilly – per habitation); **performance-based incentives (not salaried)**; **"link worker" / interface between community & health system**; drug kit (ORS, IFA, OCP, condoms, paracetamol, cotrimoxazole, zinc, chloroquine, RDT kit); **ASHA facilitator – 1 per 20 ASHAs**; induction training in modules (Modules 1–7).
- **AWW (Anganwadi Worker)** – under **ICDS (Ministry of Women & Child Development)** – **1 Anganwadi Centre per 400–800 population** (rural/ urban, plain areas).

2.6 National Health Policy (NHP)

Policy	Key points
NHP 1983	First NHP; **"Health for All by 2000"**, PHC approach, VHGs
NHP 2002	Increase public health spending to **2 % of GDP**; eradicate polio/ yaws, eliminate leprosy
NHP 2017	**Universal health coverage; "assured comprehensive care"**; public health expenditure **2.5 % of GDP by 2025**; **HWCs**; targets – **life expectancy 70 by 2025**, **TFR 2.1 by 2025**, **U5MR 23 by 2025**, **MMR 100 by 2020**, **IMR 28 by 2019**, **NMR 16 & still birth rate single digit by 2025**; TB elimination by 2025; leprosy elimination by 2018

- **National Population Policy 2000** – immediate: contraception & infrastructure; medium-term: **TFR to replacement level (2.1) by 2010**; long-term: **stable population by 2045**.
- **Health care financing** – out-of-pocket expenditure in India is high (~40–50 %; declining).

2.7 Health Team & Staffing at Sub-Centre/ PHC

Worker	Population / role
ANM / Female Health Worker (FHW)	1 per SC (5,000); **2nd ANM** under NHM; MCH, immunization, FP, records
MPW (Male) / Male Health Worker	1 per SC; malaria & vector control, sanitation, TB/ leprosy surveillance
LHV (Lady Health Visitor) / Female Health Assistant (Supervisor)	At PHC – **supervises ANMs of ~4–6 SCs**
Male Health Assistant (Supervisor)	Supervises male MPWs
CHO (Community Health Officer)	At SC-AAM – team leader for CPHC
Medical Officer (MO)	PHC in-charge
Staff Nurse	PHC (24×7 – 3 SNs), CHC (~10 SNs)
ASHA	Village (1 per 1,000)

PYQ style

- Bhore Committee – **1946**; recommended – **PHC concept, integration of curative & preventive**.
- Multipurpose worker scheme – **Kartar Singh Committee (1973)**.
- Village health guides – **Shrivastava Committee (1975)**.
- Integration of health services – **Jungalwalla Committee (1967)**.
- Alma-Ata – **1978**; goal – **Health for All by 2000**.
- PHC principles – **equitable distribution, community participation, intersectoral coordination, appropriate technology**.
- Sub-centre population (plain) – **5,000**; PHC – **30,000**; CHC – **1,20,000**.
- Hilly/ tribal SC – **3,000**; PHC – **20,000**; CHC – **80,000**.
- CHC beds – **30**; specialists – **4**; FRU – **CHC**.
- ASHA – **1 per 1,000**, age **25–45 yrs**.
- HWCs launched – **14 April 2018**; renamed – **Ayushman Arogya Mandir**.
- PM-JAY cover – **₹5 lakh per family per year**; launched – **23 Sept 2018**.
- NRHM launched – **12 April 2005**; NUHM – **1 May 2013**.
- Health in Constitution – **State subject**.
- NHP 2017 health expenditure target – **2.5 % of GDP**.

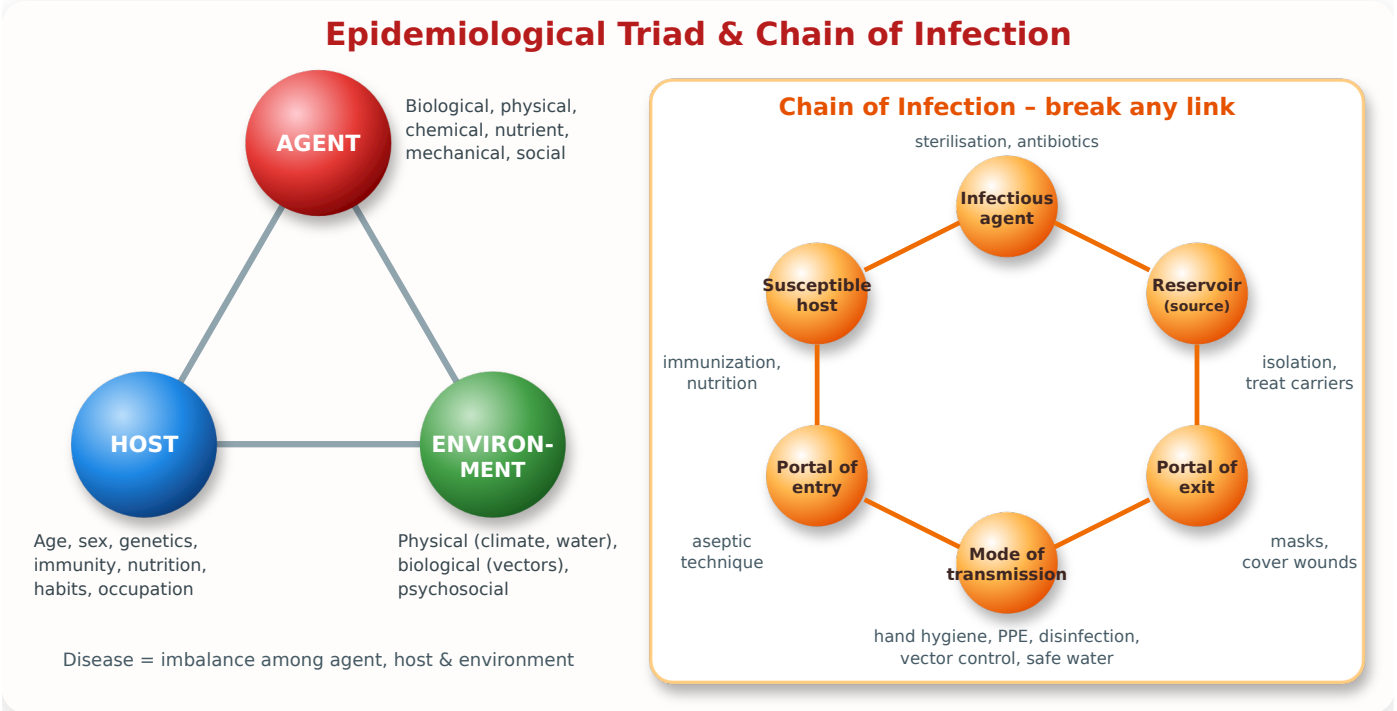
3 Epidemiology

3.1 Basics

D Epidemiology (Last, 1988)

"The study of the **distribution and determinants** of health-related states or events in specified populations, and the **application** of this study to the **control of health problems**." Greek: epi = upon, demos = people, logos = study. **Father of epidemiology – Hippocrates**; **father of modern epidemiology – John Snow** (cholera, Broad Street pump, London 1854).

- **Aims** – describe distribution & size of disease problems, identify aetiological factors, provide data for planning, implementation & evaluation of services.
- **Components** – **disease frequency, distribution (time, place, person), determinants**.
- **Epidemiological triad** – **agent, host, environment**.



Agent–host–environment triad & chain of infection

Terms

Term	Meaning
Endemic	**Constant presence** of a disease within a given geographic area (e.g. malaria, common cold)
Epidemic	Occurrence **clearly in excess of normal expectancy** in a community/ region
Pandemic	Epidemic spread **across countries/ continents** (COVID-19, influenza 1918)
Sporadic	Scattered, irregular cases (tetanus, polio now)
Outbreak	Epidemic limited to localised increase (village, hostel)
Exotic	Imported into country where it doesn't exist (rabies in UK)

Term	Meaning
Zoonosis	Transmitted between vertebrate animals & man (rabies, plague, anthrax)
Epizootic	Epidemic in animals
Nosocomial	Hospital-acquired infection
Iatrogenic	Physician/ treatment-induced
Opportunistic	Infection in immunocompromised host

3.2 Measurements in Epidemiology

Tool	Definition	Example
Rate	Event in a population over **time** × multiplier; numerator **is part of denominator**	Death rate, birth rate
Ratio	**a : b** – numerator **not part of** denominator	Sex ratio, doctor–population ratio, MMR (ratio)
Proportion	**a / (a + b)** – numerator included in denominator; usually %	% of children immunized

- **Crude rate** – entire population; **specific rate** – age/ sex/ cause specific; **standardised rate** – adjusted (direct/ indirect) for comparison.

Incidence & prevalence

- **Incidence** – number of **NEW cases** in a specified period ÷ population at risk × 1000. Measures **risk**; useful for **acute diseases & aetiology** studies.
- **Attack rate** – incidence in an epidemic (**% – cases ÷ population at risk during a limited period**).
- **Secondary attack rate (SAR)** – **(number of exposed persons developing disease within the range of incubation period following exposure to primary case) ÷ (total susceptible contacts) × 100** – **primary case excluded** from numerator & denominator; measures **infectiousness/ communicability**.
- **Prevalence** – **ALL cases (old + new)** at a point/ period ÷ population × 1000; **point prevalence** (one point in time) / **period prevalence** (incidence + point prevalence); useful for **chronic diseases**, planning services.
- **Prevalence = Incidence × Mean duration (P = I × D)**.
- **Case fatality rate (CFR)** – **deaths from a disease ÷ cases of that disease × 100** – measures **killing power/ virulence** (e.g. rabies ~100 %).
- **Proportional mortality rate** – deaths from a specific cause ÷ total deaths × 100.
- **Survival rate**, **morbidity**: frequency (incidence, prevalence) & duration.

3.3 Epidemiological Methods (Study Designs)

Type	Design	Key point
Descriptive	Time, place, person ("**who, when, where**")	**First phase**; generates hypothesis

Type	Design	Key point
Analytical – Case-control (retrospective)	**Effect → cause**; cases (disease) vs controls (no disease); look back for exposure	**Odds ratio**; quick, cheap, **rare diseases**; recall & selection bias
Analytical – Cohort (prospective/ longitudinal/ incidence/ follow-up)	**Cause → effect**; exposed vs non-exposed followed up	**Incidence, relative risk (RR), attributable risk**; expensive, long, loss to follow-up; Framingham heart study
Cross-sectional	One point in time	**Prevalence** study
Experimental – RCT (Randomised Controlled Trial)	Randomisation, intervention vs control; **blinding (single, double, triple)**	**Gold standard/ most powerful** for causality & drug efficacy
Field trials / community trials	Healthy people / whole communities (vaccine trial)	–

- **Relative risk (RR)** = incidence among exposed ÷ incidence among non-exposed (**strength of association**). **Attributable risk (AR)** = incidence exposed − non-exposed (÷ exposed × 100). **Odds ratio** – approximate RR in case-control.
- **Hierarchy of evidence** – **systematic review/ meta-analysis > RCT > cohort > case-control > cross-sectional > case series/ reports > expert opinion**.
- **Bias** – selection, information (recall, interviewer), confounding; **randomisation** removes selection bias & confounding; **double blinding** removes observer & subject bias.
- **Criteria for causation (Bradford Hill)** – **temporal association (most important)**, strength, dose–response, consistency, specificity, biological plausibility, coherence.

3.4 Screening

- **Screening** – search for unrecognised disease/ defect by rapid tests in **apparently healthy** people (**secondary prevention**). **Mass**, **high-risk/ selective**, **multiphasic (several tests at once)**, **opportunistic (case finding)**.
- **Criteria (Wilson & Jungner, 1968)** – important health problem, recognisable latent stage, natural history known, suitable & acceptable test, effective treatment available, cost-effective, continuous process.

	Disease present	Disease absent
Test +	**a – True positive**	**b – False positive**
Test −	**c – False negative**	**d – True negative**

- **Sensitivity = a/(a + c)** – ability to identify **correctly those who HAVE disease** (true positive rate). High sensitivity → few **false negatives** → **good for screening (rule out – "SnNout")**.
- **Specificity = d/(b + d)** – ability to identify correctly those **WITHOUT disease** (true negative rate). High specificity → few **false positives** → **confirmatory ("SpPin")**.
- **Positive predictive value (PPV) = a/(a + b)**; **NPV = d/(c + d)**. **PPV increases with prevalence**.
- **Validity** (accuracy – sensitivity & specificity), **reliability** (repeatability/ precision), **yield** (amount of previously unrecognised disease detected).

3.5 Investigation of an Epidemic (Steps)

1. **Verify the diagnosis** (clinical, lab).
2. **Confirm the existence of an epidemic** (compare with previous data – excess over expected).
3. **Define population at risk** (map, census).
4. **Rapid search for all cases & their characteristics** (case definition, line listing).
5. **Data analysis – time (epidemic curve), place (spot map), person**.
6. **Formulate hypothesis**.
7. **Test hypothesis**.
8. **Evaluate ecological factors**.
9. **Further investigation of population at risk**.
10. **Writing the report** + control measures (**start control measures early**, alongside).

- **Epidemic curve** – **common-source single exposure** (point source – **sharp rise & fall, all cases within one incubation period** – e.g. food poisoning, Bhopal gas tragedy), **common source continuous/ repeated**, **propagated (person-to-person – gradual rise, tails off, over several incubation periods – e.g. hepatitis A, measles)**, slow (modern) epidemics (CHD, cancer).
- **Spot map** – John Snow's cholera map.

PYQ style

- Father of modern epidemiology – **John Snow**.
- Constant presence of disease in an area – **endemic**; across countries – **pandemic**.
- Disease transmitted from animals to man – **zoonosis**.
- Rate vs ratio – in **ratio, numerator not part of denominator**.
- New cases – **incidence**; all cases – **prevalence**; **P = I × D**.
- Killing power of disease – **case fatality rate**.
- Infectiousness measured by – **secondary attack rate**.
- Case-control gives – **odds ratio**; cohort gives – **relative risk**.
- Gold standard study – **RCT**; removes bias – **randomisation & blinding**.
- Most important causation criterion – **temporal association**.
- Sensitivity – **true positives / all diseased**; specificity – **true negatives / all non-diseased**.
- Screening is – **secondary prevention**.
- First step in epidemic investigation – **verify diagnosis**.
- Point-source epidemic – **food poisoning (all within one incubation period)**.

4 Biostatistics, Demography & Vital Statistics

4.1 Biostatistics – Basics

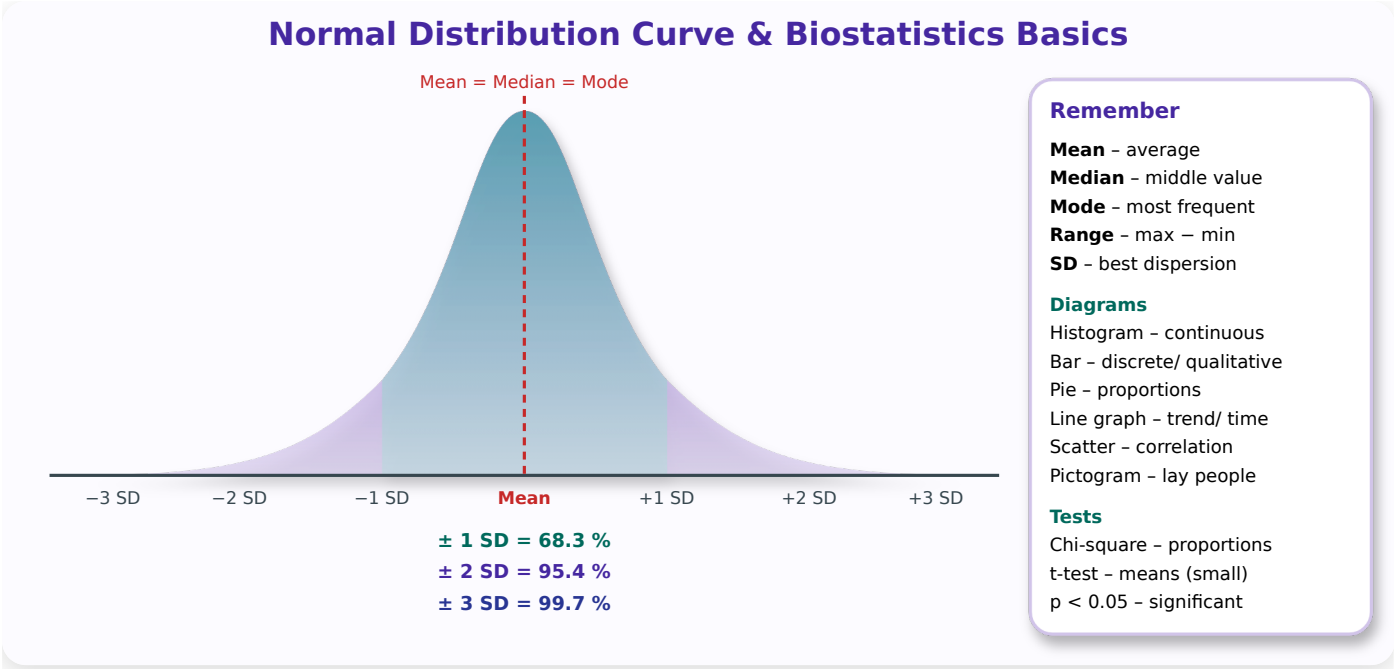
- **Biostatistics** – application of statistical methods to biological/ medical data. **Father of biostatistics – Francis Galton** (statistics in general – Sir Ronald Fisher is father of modern statistics).
- **Data types** – **qualitative (categorical – nominal: sex, blood group; ordinal: ranked – grades of pain)** & **quantitative (numerical – discrete: number of children; continuous: Hb, height, weight)**.
- **Sources of data** – **census**, **registration of vital events (CRS)**, **SRS (Sample Registration System)**, notification of diseases, hospital records, **NFHS, surveys**, **HMIS**, disease registers.

Presentation of data

Method	Use
Tabulation	Simple/ complex tables
Bar diagram	**Discrete/ qualitative** data comparison (simple, multiple, component/ proportional)
Histogram	**Continuous quantitative** data (frequency distribution) – no gaps between bars
Frequency polygon	Joining midpoints of histogram
Line diagram/ graph	**Trend over time**
Pie chart	**Proportions/ percentages** of a whole (angle = % × 3.6°)
Pictogram	**Lay people** – pictures
Scatter diagram	**Correlation** between two variables
Spot map	Geographic distribution (John Snow)
Ogive	Cumulative frequency (median from graph)

Central tendency & dispersion

- **Mean** (arithmetic average) – affected by extreme values. **Median** – middle value of arranged data (best when skewed). **Mode** – most frequent value. **In normal distribution – mean = median = mode**.
- **Range** – highest − lowest (simplest); **mean deviation**; **standard deviation (SD) – most commonly used & best measure of dispersion** – $\sqrt{[\Sigma(x - \bar{x})^2 / (n - 1)]}$; **variance = SD²**; **coefficient of variation = SD/ mean × 100**.
- **Example**: values 2, 4, 4, 5, 10 → mean = 25/5 = **5**, median = **4**, mode = **4**, range = **8**.



Normal distribution & key biostatistics points

- **Normal (Gaussian) curve** – bell-shaped, symmetrical; **mean ± 1 SD = 68.27 %, ± 2 SD = 95.45 %, ± 3 SD = 99.73 %**; **mean ± 1.96 SD = 95 %**.
- **Skewness** – positive (tail right; mean > median > mode), negative (tail left).
- **Standard error (SE) = SD/√n**.

Sampling

- **Probability (random)** – **simple random** (lottery, random number tables), **systematic** (every nth), **stratified** (sub-groups – heterogeneous population), **cluster** (e.g. **30-cluster survey – immunization coverage (WHO)**), multistage.
- **Non-probability** – convenience, purposive, quota, snowball.
- **Sampling error** – reduced by increasing sample size.

Tests of significance

- **Null hypothesis (H_0)** – no difference; **$p < 0.05$** → significant (reject H_0).
- **Type I error (α)** – rejecting a true H_0 (false positive); **Type II error (β)** – accepting a false H_0; **power = 1 − β**.
- **Chi-square test** – **qualitative data/ proportions**; **t-test (Student's)** – **difference between two means (small samples < 30)**; paired t-test (before–after); **Z-test** – large samples; **ANOVA** – > 2 means; **correlation coefficient (r)** – −1 to +1.

4.2 Demography

- **Demography** – scientific study of human population – **size, composition (age, sex), distribution** & changes (fertility, mortality, migration). **Father of demography – John Graunt**.
- **Demographic cycle (5 stages)**: (1) **High stationary** (high birth & death rates – India till 1920); (2) **Early expanding** (death rate falls, birth high); (3) **Late expanding** (death rate falls further, birth rate starts falling – **India now**); (4) **Low stationary** (both low – Europe); (5) **Declining** (birth < death – Germany, Japan).

- **Population pyramid** – **broad base (high fertility – developing, India's earlier)** vs narrow base/ bulging top (developed, ageing).
- **Dependency ratio** – (population < 15 + > 64) ÷ population 15–64 × 100. **Demographic dividend** – ↑ working-age share.

Census

- **Census of India** – every **10 years**; **first (non-synchronous) 1872 (Lord Mayo)**; **first synchronous 1881 (Lord Ripon)**; conducted by **Registrar General & Census Commissioner (MHA)**; **Census 2011 – 15th**; next census postponed (COVID) – **Census 2027** (scheduled, includes caste enumeration).

Census 2011 (final)	Value
Population	**121.09 crore (1.21 billion)**
Decadal growth (2001–11)	**17.7 %**
Sex ratio (females per 1000 males)	**943** (Kerala highest 1084; Haryana lowest 879)
Child sex ratio (0–6 yrs)	**919**
Literacy	**74.04 %** (male 82.1, female 65.5); Kerala highest
Density	**382 per km²** (Bihar highest)
Gujarat	Population ~6.04 crore; sex ratio **919**; literacy **78.0 %**

- **NFHS-5 (2019–21)** – **TFR 2.0** (below replacement 2.1); **sex ratio 1020** (first time > 1000); stunting 35.5 %, wasting 19.3 %, underweight 32.1 %; anaemia – children 6–59 m **67.1 %**, women 15–49 **57.0 %**; institutional births 88.6 %; full immunization 76.4 %.

4.3 Vital Statistics & Registration

- **Registration of Births & Deaths (RBD) Act 1969** – **compulsory registration**; report **birth & death within 21 days** (late fee after); **still birth** also registered; **Registrar** at village – Panchayat secretary/ (urban – municipal health officer); **RBD Amendment Act 2023** – **digital birth certificate as single document for date of birth**, national database.
- **CRS (Civil Registration System)**; **SRS (Sample Registration System, 1964–65 pilot, full 1969–70)** – **dual record system** (continuous enumeration by part-time enumerator + independent 6-monthly survey by supervisor) – gives reliable **annual birth & death rates, IMR, MMR, TFR**.
- **MCCD (Medical Certification of Cause of Death)** – ICD-10/ ICD-11 coding.

Fertility & mortality indicators

Indicator	Formula	India (latest SRS)
Crude Birth Rate (CBR)	Live births ÷ **mid-year population** × 1000	**~19.3 (2021)**
Crude Death Rate (CDR)	Deaths ÷ mid-year population × 1000	**~7.5 (2021)**
General Fertility Rate (GFR)	Live births ÷ **women 15–49 yrs** × 1000	–
Total Fertility Rate (TFR)	Average **number of children a woman would bear** in her reproductive life at current rates	**2.0** (SRS 2021/ NFHS-5)

Indicator	Formula	India (latest SRS)
Gross Reproduction Rate (GRR)	Average number of **daughters** a woman would have	–
Net Reproduction Rate (NRR)	Daughters, **adjusting for mortality**; **NRR = 1 ≈ TFR 2.1 (replacement level)**	~1
IMR	Infant deaths (< 1 yr) ÷ **live births** × 1000	**27 (2021)**
NMR	Neonatal deaths (< 28 days) ÷ live births × 1000	**19**
Post-neonatal MR	28 days – 1 yr deaths ÷ live births × 1000	–
Perinatal MR	(Still births ≥ 28 wks + early neonatal deaths) ÷ **(live births + still births)** × 1000	–
U5MR	Under-5 deaths ÷ live births × 1000	**31**
MMR (ratio)	Maternal deaths ÷ **live births** × **1,00,000**	**93 (2019–21)**
Maternal mortality rate	Maternal deaths ÷ **women 15–49** × 1,00,000	–
Life expectancy at birth	–	**~70 yrs**
Growth rate	CBR − CDR (natural increase)	~1.2 %
Couple Protection Rate (CPR)	% eligible couples effectively protected by any contraceptive	NFHS-5 modern method use ~56.5 %

- **Eligible couple** – currently married couple, **wife aged 15–49 yrs**; **eligible couple register** maintained by ANM (updated yearly – April).
- **Kerala** – lowest IMR & MMR; highest life expectancy.

PYQ style

- Best measure of dispersion – **standard deviation**.
- Mean = median = mode – **normal distribution**.
- Mean ± 2 SD covers – **95.4 %**.
- Trend over time – **line diagram**; continuous data – **histogram**; proportions – **pie chart**; correlation – **scatter diagram**.
- Immunization coverage survey – **30-cluster sampling**.
- Chi-square test – **qualitative data/ proportions**; t-test – **means of small samples**.
- Father of demography – **John Graunt**.
- India in demographic cycle – **late expanding stage**.
- First synchronous census – **1881**; census interval – **10 years**.
- Census 2011 sex ratio – **943**; child sex ratio – **919**; literacy – **74.04 %**.
- Registration of birth/ death within – **21 days (RBD Act 1969)**.
- SRS uses – **dual record system**.
- CBR denominator – **mid-year population**; GFR denominator – **women 15–49**.
- Replacement level fertility – **TFR 2.1 / NRR 1**.
- India TFR (NFHS-5) – **2.0**.

5 Communicable Disease Epidemiology & Control

5.1 Basic Terms

Term	Meaning
Infection	Entry & development/ multiplication of an infectious agent in man/ animal
Contamination	Presence of agent on body surface/ articles (not multiplication)
Infestation	Lodgement & development of **arthropods** on body surface/ clothing (lice, scabies)
Source	Person/ animal/ object from which agent passes **immediately** to host
Reservoir	Where agent **normally lives & multiplies** (human, animal, non-living – soil)
Carrier	Infected person who **harbours the agent without clinical disease & serves as source** – **incubatory** (measles, mumps, polio, diphtheria, pertussis), **convalescent** (typhoid, diphtheria, dysentery, cholera), **healthy** (polio, cholera, diphtheria, salmonella), **temporary** (< 6 months) vs **chronic** (> 6 months – **typhoid (gallbladder), hepatitis B**, malaria, gonorrhoea)
Incubation period	Time from entry of agent to onset of first symptom
Generation time	Interval from receipt of infection to **maximal infectivity** of host
Serial interval	Gap between onset of primary & secondary cases
Communicable period	Time during which agent may be transferred
Herd immunity	Level of resistance of a community/ group to an infection (immunized majority protects others) – **not applicable to tetanus**

- **Incubation period** importance – tracing source, **period of surveillance/ quarantine (= max IP)**, immunization of contacts, identify point-source vs propagated.
- **Short IP** – **cholera (1–2 days), food poisoning (hours), influenza (1–3 days)**, diphtheria (2–6 days). **Long IP** – **hepatitis B (45–180 days), leprosy (years), rabies (days–years; usually 1–3 months)**.

5.2 Modes of Transmission

Direct	Indirect
Direct contact (STIs, leprosy, skin infections, scabies)	**Vehicle-borne** – water, food, milk, blood, fomites… (cholera, typhoid, hepatitis A, polio, hepatitis B via blood)
Droplet infection (< 1 m – common cold, diphtheria, pertussis, TB, influenza, meningococcal)	**Vector-borne** – **mechanical** (housefly – typhoid, cholera) or **biological** (mosquito – malaria)
Contact with soil (tetanus, hookworm)	**Airborne** – **droplet nuclei** (TB, measles, chickenpox – long distance) & **dust**
Inoculation into skin/ mucosa (rabies, hepatitis B)	**Fomite-borne** – articles (diphtheria, typhoid, HAV)
Transplacental (vertical) – TORCH, HIV, syphilis, HBV	**Unclean hands & fingers** – most common in faeco-oral

- **Biological transmission types** – **propagative** (multiply, no development – plague in rat flea), **cyclo-developmental** (develop, no multiply – **filaria in mosquito**), **cyclo-propagative** (both – **malaria in Anopheles**).
- **Extrinsic incubation period** – time in vector (malaria 10–14 days, dengue 8–10 days).

5.3 Control of Communicable Diseases

(A) Control of reservoir

- **Early diagnosis**, **notification**, **epidemiological investigation**, **isolation**, **treatment**, **quarantine**, disinfection.
- **Isolation** – **separation of infected persons** for the **period of communicability** (in hospital/ home). Types – strict, respiratory (droplet/ airborne), contact, enteric, blood-body fluid; **standard precautions** for all. **Not useful** where many subclinical cases (polio, hepatitis A).
- **Quarantine** – limitation of freedom of movement of **well (healthy) contacts** exposed to communicable disease for a period **not longer than the longest incubation period**; e.g. **yellow fever – 6 days**; **absolute** (complete), **modified** (selective – e.g. exclude children from school), **segregation**. **International Health Regulations (IHR 2005)**.
- **Surveillance** – continuous scrutiny; **active** (health worker goes house to house – **malaria fever cases, AFP**) vs **passive** (cases come to facility); **sentinel surveillance** (selected sites – HIV); **syndromic**.
- **Notifiable diseases** – reporting to health authority mandatory; international (IHR) – **cholera, plague, yellow fever** (+ SARS, smallpox, new influenza subtype, wild polio).

(B) Interrupt transmission

- Water & food sanitation, **hand washing (most effective single measure in hospital infection control)**, vector control (environmental, chemical, biological – **Gambusia fish, BTI**), disinfection, safe sex, safe blood.

(C) Protect susceptible host

- **Active immunization**, **passive immunization (immunoglobulins)**, **chemoprophylaxis** (e.g. INH for TB contacts, rifampicin for meningococcal contacts, doxycycline for leptospirosis, SDR-PEP for leprosy), non-specific measures (nutrition, health education, personal hygiene).

5.4 Disinfection

Term	Meaning
Disinfection	Killing infectious agents **outside the body** by chemical/ physical means (**not spores**)
Concurrent disinfection	**Immediately after discharge** of infectious material from body (stool, urine, sputum, soiled linen) – during illness
Terminal disinfection	After patient has **left/ died/ isolation ends** – room, bed, articles (largely replaced by **terminal cleaning**)
Precurrent (prophylactic)	**Before** – chlorination of water, pasteurisation of milk, hand washing
Sterilisation	Destruction of **all forms including spores**

Term	Meaning
Antiseptic	Inhibits/ kills organisms on **living tissue**
Deodorant	Removes odour

Agent	Use
Bleaching powder (33 % chlorine)	Water, faeces, urine, floors
Sodium hypochlorite 1 % (0.5 % surfaces)	**Blood spills, BMW, HBV/ HIV contaminated** surfaces
Phenol (carbolic acid 5 %)	**Standard disinfectant – Rideal-Walker coefficient** (phenol coefficient)
Cresol (Lysol 2–5 %)	Faeces, sputum, floors
Alcohol 70 %	Skin, thermometers, hand rub (60–80 %)
Formaldehyde (formalin)	Fumigation of rooms (OT), preservation
Glutaraldehyde 2 % (Cidex)	**Endoscopes, heat-sensitive instruments** (high-level disinfectant; sterilant in 10 hrs)
Chlorhexidine	Skin, hand wash, cord care
Povidone iodine	Skin preparation
Lime (quick lime)	Faeces, burial
Burning/ incineration	Contaminated dressings, anatomical waste
Autoclave 121 °C, 15 lb/in², 15–20 min	Linen, instruments – **sterilisation**
Hot air oven 160 °C × 2 hrs	**Glassware, powders, oils**
Boiling (100 °C, 10–30 min)	Not sterilisation (spores survive)
Pasteurisation – **holder 63–66 °C × 30 min**; **flash/ HTST 72 °C × 15–20 sec**	Milk

- **Spaulding classification** – **critical** (enter sterile tissue – sterilise), **semi-critical** (mucous membranes – high-level disinfection), **non-critical** (intact skin – low-level).

5.5 Surveillance – IDSP/ IHIP

- **IDSP (Integrated Disease Surveillance Project/ Programme)** – launched **Nov 2004**; **3 levels of case definitions**: **S – Syndromic** (by health worker, **S form**), **P – Presumptive** (by MO, **P form**), **L – Laboratory confirmed** (**L form**); weekly reporting (Monday–Sunday); **Rapid Response Team (RRT)** at district; **IHIP (Integrated Health Information Platform)** – digital, real-time (2021); **call centre 1075**.
- **AFP surveillance** – < 15 yrs; **measles–rubella surveillance (fever + rash)**; **Ni-kshay** (TB); **HMIS**; **IHIP**.
- **Epidemic threshold**; **early warning signals** – clustering of cases, unusual deaths.

PYQ style

- Person harbouring agent without disease – **carrier**; chronic carrier – **typhoid (gallbladder)**.
- Infestation – **arthropods on skin/ clothes (lice, scabies)**.
- Period of quarantine – **longest incubation period**; yellow fever – **6 days**.
- Isolation is for – **infected persons**; quarantine – **healthy contacts**.
- Malaria in mosquito – **cyclo-propagative**; filaria – **cyclo-developmental**.
- Most effective measure against hospital infection – **hand washing**.
- Disinfection immediately after discharge of infectious material – **concurrent**.
- After patient leaves – **terminal disinfection**.
- Standard for comparing disinfectants – **phenol (Rideal-Walker coefficient)**.
- Blood spill disinfectant – **1 % sodium hypochlorite**.
- Endoscope disinfection – **2 % glutaraldehyde**.
- Autoclave – **121 °C, 15 lb, 15–20 min**; hot air oven – **160 °C, 2 hrs**.
- Flash pasteurisation – **72 °C × 15–20 sec**.
- IDSP launched – **2004**; forms – **S, P, L**.
- International notifiable (IHR) – **cholera, plague, yellow fever**.

6 National Programmes – Communicable Diseases

6.1 Tuberculosis – NTEP

- **Mycobacterium tuberculosis** (Robert Koch, **24 March 1882 – World TB Day**); **droplet nuclei (airborne)**; **one untreated sputum +ve patient infects 10–15 people/ year**; India has **~26–27 % of global TB burden** (highest).
- **Programme history** – **NTP 1962** → **RNTCP 1997 (DOTS)** → **NTEP (National TB Elimination Programme) – 1 Jan 2020**; **National Strategic Plan 2017–25 – "End TB by 2025"** (India's target; SDG 2030); **Pradhan Mantri TB Mukt Bharat Abhiyaan (Sept 2022)** – **Ni-kshay Mitra** (community support/ food baskets); **100-day intensified campaign (Dec 2024)**.
- **Presumptive TB** – **cough ≥ 2 weeks**, fever, weight loss, night sweats, haemoptysis, any abnormal CXR; **contacts, PLHIV – any duration cough**.
- **Diagnosis** – **NAAT (CBNAAT/ GeneXpert, Truenat) – upfront for all** (also detects **rifampicin resistance**); sputum smear microscopy (**ZN stain**, LED fluorescence); **2 sputum samples** (one spot, one early morning); CXR; culture (gold standard – LJ medium/ MGIT).
- **Treatment (drug-sensitive)** – **Intensive phase 2 months HRZE + continuation phase 4 months HRE (daily, fixed-dose combination, weight bands)**; **6 months total**; extended continuation in TBM/ bone TB.
- **Drug side effects** – **INH: peripheral neuropathy (give pyridoxine), hepatitis**; **Rifampicin: orange-red urine/ tears, hepatitis, drug interactions (OCP failure)**, flu-like; **Pyrazinamide: hepatotoxicity, hyperuricaemia (gout)**, arthralgia; **Ethambutol: optic neuritis (red-green colour blindness)**; **Streptomycin: ototoxicity, nephrotoxicity**.
- **DR-TB** – **MDR-TB (resistant to H & R)**; **XDR**; **shorter oral bedaquiline-containing regimen / BPaLM (bedaquiline, pretomanid, linezolid, moxifloxacin – 6 months)**.
- **TPT (TB preventive therapy)** – household contacts (all ages, after ruling out active TB), PLHIV – **6H (INH 6 months)** or **3HP (INH + rifapentine weekly × 12 weeks)**.
- **Ni-kshay Poshan Yojana (2018)** – **DBT ₹500/month** → **enhanced to ₹1,000/month (Nov 2024)** during treatment; **Ni-kshay portal** (case notification – **mandatory notification of TB (2012)**, including private sector).
- **Nurse role** – case finding, **DOT (Directly Observed Treatment) / treatment supporter**, adherence (**99DOTS, MERM**), contact tracing, **cough etiquette**, sputum collection (open air, deep cough, 2–5 ml), nutrition, follow-up sputum at end of IP & treatment.
- **BCG** – protects children against severe TB (TBM, miliary).

6.2 Leprosy – NLEP

- **Mycobacterium leprae** (**Hansen, 1873**); **longest incubation (avg 3–5 yrs)**; **droplet (nasal discharge) – main route**; affects **skin & peripheral nerves**; **not highly infectious**; **lepromin test** (prognostic, not diagnostic).
- **Cardinal signs (any one = leprosy)**: (1) **hypopigmented/ reddish skin patch with definite loss of sensation**, (2) **thickened/ enlarged peripheral nerve with loss of sensation/ weakness** of muscles supplied, (3) **AFB in slit-skin smear**.

- **Nerves commonly thickened** – **ulnar (most common)**, common peroneal (lateral popliteal), posterior tibial, great auricular, radial cutaneous, median, facial.
- **Classification (WHO – for treatment)**: **PB (paucibacillary) – 1–5 skin lesions, no nerve / only 1 nerve, smear negative**; **MB (multibacillary) – ≥ 6 lesions / > 1 nerve / smear positive**.
- **MDT (Multi-Drug Therapy – 1982 WHO; India 1983)** – **rifampicin (monthly supervised) + clofazimine + dapsone**; **PB – 6 months**, **MB – 12 months**; (WHO now recommends the **3-drug regimen for PB as well**); blister packs; **rifampicin – red urine; clofazimine – skin pigmentation (reddish-black), dryness; dapsone – haemolysis (G6PD), DRESS**.
- **Reactions** – **Type 1 (reversal)** – existing lesions inflamed, **neuritis** → **prednisolone**; **Type 2 (ENL – erythema nodosum leprosum)** – MB, painful nodules, fever → prednisolone, **thalidomide**, clofazimine.
- **Deformities** – **claw hand (ulnar), foot drop (peroneal), lagophthalmos (facial), plantar ulcers**; **self-care (soak, scrape, oil)**, **MCR footwear**, reconstructive surgery (**RCS**) – incentive.
- **Elimination as public health problem – prevalence < 1 per 10,000** (national level achieved **Dec 2005**); **SDR-PEP (single-dose rifampicin post-exposure prophylaxis)** to contacts; **LCDC (Leprosy Case Detection Campaign)**; **SPARSH Leprosy Awareness Campaign (30 Jan – Anti-Leprosy Day, Gandhiji's death anniversary)**; **National Strategic Plan – zero transmission by 2030**.

6.3 Vector-Borne Diseases – NCVBDC

- **NCVBDC (National Center for Vector Borne Diseases Control)** (earlier **NVBDCP – 2003/04**; NMEP 1958; NMCP 1953) – **6 diseases: malaria, dengue, chikungunya, Japanese encephalitis, kala-azar, lymphatic filariasis**.

Disease	Vector	Key points
Malaria	**Female Anopheles** (night biter; **An. culicifacies** – rural, **An. stephensi** – urban)	**P. vivax (most common), P. falciparum (most dangerous – cerebral)**; **RDT/ blood smear (thick & thin)**; vivax – **chloroquine 3 days + primaquine 14 days**; falciparum – **ACT (AS + SP; AL in NE states) + primaquine single dose**; severe – **injectable artesunate**; **LLINs, IRS (indoor residual spraying)**, larvivorous fish; **NFME 2016 – malaria elimination by 2030**; **ABER (annual blood examination rate) ≥ 10 %**; **API (annual parasite incidence) – positives per 1000 population**; **World Malaria Day 25 April**
Dengue	**Aedes aegypti** (day biter, **breeds in clean stored water** – coolers, tyres, pots)	Fever, myalgia, rash, thrombocytopenia; **NS1 (early)/ IgM ELISA**; **no NSAIDs/ aspirin**; **source reduction – "dry day" weekly**; **National Dengue Day – 16 May**
Chikungunya	Aedes aegypti/ albopictus	Fever + **severe polyarthralgia** ("bends up")
Japanese encephalitis (JE)	**Culex tritaeniorhynchus/ vishnui** (rice fields); **pigs – amplifier**, ardeid birds – reservoir	AES; **JE vaccine (2 doses – 9 m & 16–24 m) in endemic districts**; man – dead-end host
Lymphatic filariasis	**Culex quinquefasciatus** (dirty water); **Wuchereria bancrofti** (most), **Brugia malayi** (Mansonia)	**Nocturnal periodicity – night blood smear (8:30 PM – 12 midnight / 10 PM–2 AM)**; lymphoedema, **hydrocele, elephantiasis**; **MDA – DEC + albendazole (DA)** or **IDA (+ ivermectin)** annually; **morbidity management (MMDP)** – limb hygiene, hydrocele surgery; target elimination

Disease	Vector	Key points
Kala-azar (visceral leishmaniasis)	**Sandfly – Phlebotomus argentipes** (cracks in mud walls)	**Leishmania donovani**; Bihar, Jharkhand, UP, WB; **fever, massive splenomegaly, darkening of skin, pancytopenia**; **rK39 RDT**; **single-dose liposomal amphotericin B**; **IRS (synthetic pyrethroid)**; **elimination target < 1 case/10,000 at block level – achieved in all endemic blocks (2023)**; **PKDL** – post kala-azar dermal leishmaniasis

- **Integrated vector management** – **source reduction (most effective, cheapest)**, **larvivorous fish – Gambusia, Guppy**, **temephos (Abate) – larvicide in drinking water**, **BTI**, fogging (adult control, outbreaks), personal protection (repellents, nets, full sleeves).

6.4 HIV/ AIDS – NACP

- **NACO (National AIDS Control Organisation) – 1992**; **NACP phase I 1992 → phase V (2021–26)**; **goal – end AIDS as public health threat by 2030**; **95-95-95 targets** (95 % know status, 95 % of them on ART, 95 % of them virally suppressed).
- **Transmission** – **sexual (most common in India ~85–90 %)**, **parent-to-child**, blood, needle sharing (IDUs). **Highest risk per exposure – blood transfusion (~90 %)**.
- **Window period** – ~3–12 weeks (antibody tests negative though infectious).
- **Diagnosis (NACO)** – **3 different rapid tests (E/R/S – different antigens/ principles)** for diagnosis in adults; infants – **DNA PCR (EID) at 6 weeks**; **ICTC (Integrated Counselling & Testing Centre)** – pre-test & post-test counselling, **informed consent & confidentiality**.
- **"Test & Treat" policy (2017)** – ART for all PLHIV regardless of CD4; **first-line – TLD (tenofovir + lamivudine + dolutegravir)**; viral load monitoring.
- **PPTCT** – universal screening of pregnant women; lifelong ART; infant **nevirapine prophylaxis**.
- **Targeted interventions (TI)** for **high-risk groups** – FSW, MSM, IDU, transgender, migrants, truckers; **OST (opioid substitution)**; condom promotion; **STI clinics – "Suraksha" (colour-coded kits)**; **blood safety – NAT-tested, voluntary donors**.
- **HIV/ AIDS (Prevention & Control) Act 2017** – **prohibits discrimination**, informed consent, confidentiality; **Ombudsman** in each state.
- **World AIDS Day – 1 December**; **Red ribbon**; **toll-free 1097**.
- **PEP (post-exposure prophylaxis)** – start **within 2 hrs (best), ≤ 72 hrs**, **28 days** of ART (TLD); wash wound with soap & water (**don't squeeze/ scrub**).

6.5 Other Communicable Disease Programmes

Disease/ Programme	Key points
Polio (Pulse Polio – 1995)	**India polio-free – 27 March 2014** (last case 13 Jan 2011, Howrah); **NIDs/ SNIDs**; **AFP surveillance < 15 yrs**; bOPV + fIPV in UIP; **cVDPV** risk; National Immunization Day
National Viral Hepatitis Control Programme (NVHCP)	**28 July 2018 (World Hepatitis Day)**; **eliminate hepatitis C & reduce HBV by 2030**; free diagnosis & treatment; Hep B birth dose; HAV & HEV – faeco-oral (HEV – fatal in pregnancy)

Disease/ Programme	Key points
Rabies – NRCP (National Rabies Control Programme)	**Dog bite – 96 %**; **"Zero by 30"** (zero human rabies deaths by 2030); **wound wash 15 min with soap & running water** (first aid); **category I** (touch/ feed/ lick intact skin – no PEP), **II** (minor scratches without bleeding/ nibbling – **vaccine**), **III** (transdermal bites/ licks on broken skin/ mucosa/ bat – **vaccine + RIG**); ARV **IM (Essen – days 0, 3, 7, 14, 28)** or **ID (Updated Thai Red Cross – 0.1 ml at 2 sites on days 0, 3, 7, 28)**; **World Rabies Day – 28 Sept** (Louis Pasteur)
Cholera	**Vibrio cholerae O1/ O139**; rice-water stools; **ORS mainstay**; **doxycycline single dose (adults)**; chemoprophylaxis not recommended for mass
Typhoid	Salmonella Typhi; faeco-oral; **Widal/ blood culture**; **typhoid conjugate vaccine (TCV)**; carriers
COVID-19	SARS-CoV-2; first case in India – **30 Jan 2020 (Kerala)**; WHO pandemic declared **11 March 2020**; **vaccination started 16 Jan 2021 (Covishield, Covaxin)**; CoWIN
Antimicrobial resistance (AMR)	**National Action Plan on AMR (2017)**; **Red Line campaign** (antibiotics with red vertical line – Rx only); Kerala KARSAP
Leptospirosis, scrub typhus, Nipah, KFD, plague	Emerging/ re-emerging zoonoses; **One Health** approach

PYQ style

- World TB Day – **24 March**; RNTCP started – **1997**; renamed NTEP – **2020**.
- India's TB elimination target – **2025**; SDG – **2030**.
- TB upfront diagnostic test – **NAAT (CBNAAT/ Truenat)**.
- Drug-sensitive TB regimen – **2HRZE + 4HRE**.
- INH side effect – **peripheral neuropathy (give pyridoxine)**; ethambutol – **optic neuritis**; rifampicin – **orange urine**.
- Ni-kshay Poshan Yojana – **₹1000/month (2024)**.
- Leprosy commonest thickened nerve – **ulnar**; PB – **≤ 5 lesions**; MB treatment – **12 months**.
- Clofazimine side effect – **skin pigmentation**; Anti-Leprosy Day – **30 January**.
- Leprosy elimination – **prevalence < 1/10,000**.
- NCVBDC covers – **6 vector-borne diseases**.
- Dengue vector – **Aedes aegypti**; breeds in – **clean stored water**.
- Filaria vector – **Culex**; blood smear – **at night**; MDA – **DEC + albendazole**.
- Kala-azar vector – **sandfly**; test – **rK39**.
- JE amplifier host – **pigs**.
- Larvivorous fish – **Gambusia, Guppy**; larvicide in drinking water – **temephos (Abate)**.
- NACO established – **1992**; HIV targets – **95-95-95**; World AIDS Day – **1 Dec**.
- PEP within – **72 hrs (best 2 hrs), for 28 days**.
- Rabies category III – **vaccine + RIG**; World Rabies Day – **28 Sept**.

7 National Programmes – NCDs & Others

7.1 Non-Communicable Diseases – Overview

- **NCDs cause ~63–65 % of deaths in India** – **cardiovascular diseases (leading)**, cancers, chronic respiratory disease, diabetes.
- **Common modifiable risk factors** – **tobacco, alcohol, unhealthy diet (salt, sugar, fat), physical inactivity**, air pollution → obesity, ↑ BP, ↑ sugar, ↑ lipids.
- **WHO "25 by 25"** – 25 % reduction in premature NCD mortality by 2025; **SDG 3.4 – one-third reduction by 2030**.

7.2 NP-NCD (National Programme for Prevention & Control of NCDs)

- **NPCDCS** (Cancer, Diabetes, CVD & Stroke) – **2010** → **renamed NP-NCD (2023)**.
- **Population-Based Screening (PBS)** – **all persons ≥ 30 yrs** at **AAM** for **hypertension, diabetes, oral cancer, breast cancer, cervical cancer** (+ COPD, CKD, NAFLD added); **CBAC form** (Community-Based Assessment Checklist – ASHA fills; score > 4 → priority).
- **Screening methods** – **BP** (≥ 140/90), **random blood sugar (glucometer – ≥ 140 mg/dl → refer)**, **oral visual examination (OVE)**, **clinical breast examination (CBE)** (women ≥ 30), **VIA (visual inspection with 3–5 % acetic acid)** – women **30–65 yrs, every 5 yrs**.
- **NCD clinics** at district hospital & CHC; **"75/25 initiative" (2023)** – **75 million people with HTN/ diabetes on standard care by 2025**; **India Hypertension Control Initiative (IHCI)**.
- **Cancer** – **commonest in Indian men – lip/ oral cavity & lung**; **women – breast, then cervix**; **tertiary cancer care centres**, **Tata Memorial**; **HPV vaccine (9–14 yr girls)** being introduced.

7.3 National Programme for Control of Blindness & Visual Impairment (NPCBVI)

- **Launched 1976** (100 % centrally sponsored); **cataract – leading cause of blindness** in India; others – refractive errors (leading cause of visual impairment), glaucoma, corneal opacity, diabetic retinopathy.
- **Blindness (WHO)** – **presenting visual acuity < 3/60** in better eye; **visual impairment < 6/18**.
- Activities – **free cataract surgery (IOL)**, **school eye screening** (free spectacles), **eye donation** (**Eye Donation Fortnight 25 Aug – 8 Sept**), **eye banks**, diabetic retinopathy & glaucoma management, **trachoma eliminated (2024)**.
- **Vitamin A** – prevents nutritional blindness.
- **World Sight Day** – 2nd Thursday of October.

7.4 National Mental Health Programme (NMHP)

- **NMHP – 1982**; **DMHP (District Mental Health Programme) – 1996**; **Mental Healthcare Act 2017** (rights-based, **decriminalised suicide attempt – Section 115**, advance directives, nominated representative, **no ECT without anaesthesia; no ECT in minors**); **National Suicide Prevention**

Strategy (2022); **Tele-MANAS (Oct 2022) – toll-free 14416 / 1-800-891-4416**; **Kiran helpline**; **World Mental Health Day – 10 October**.

- **Rights of Persons with Disabilities (RPwD) Act 2016 – 21 disabilities** (incl. mental illness, autism, intellectual disability, acid attack, thalassaemia, haemophilia, sickle cell, Parkinson's); **benchmark disability ≥ 40 %**; reservation **4 % in govt jobs, 5 % in higher education**; **UDID card**.

7.5 Tobacco Control – NTCP & COTPA

- **NTCP – 2007–08**; **COTPA (Cigarettes & Other Tobacco Products Act) – 2003**:
- **Section 4 – ban on smoking in public places** (since 2 Oct 2008).
- **Section 5 – ban on advertisement**.
- **Section 6(a) – no sale to persons < 18 yrs**; **6(b) – no sale within 100 yards of educational institutions**.
- **Section 7 – pictorial health warnings (85 % of pack surface)**.
- **Tobacco Quitline – 1800-112-356**; **mCessation (SMS)**; **Prohibition of Electronic Cigarettes Act 2019** (e-cigarettes banned); **WHO FCTC (2003; India ratified 2004)**; **World No Tobacco Day – 31 May**; **GATS-2 (2016–17) – 28.6 % adults use tobacco** (smokeless more common).
- **Tobacco Free Educational Institutions (ToFEI)** guidelines.

7.6 Other Programmes

Programme	Year	Key points
NPHCE (National Programme for Health Care of the Elderly)	**2010**	**Elderly ≥ 60 yrs**; geriatric clinics at CHC/ DH, **Regional Geriatric Centres**, **National Centres for Ageing (AIIMS Delhi, MMC Chennai)**; home-based care; **Maintenance & Welfare of Parents & Senior Citizens Act 2007**; **Elder line 14567**; **International Day of Older Persons – 1 October**
NPPCD (Prevention & Control of Deafness)	**2006**	Early detection (newborn hearing screening – OAE), hearing aids; **noise > 85 dB → hearing loss**; **World Hearing Day – 3 March**
NIDDCP (National Iodine Deficiency Disorders Control Programme)	**1992** (earlier **NGCP 1962**)	**Iodised salt – 30 ppm at production, 15 ppm at consumer/ household**; IDD – goitre, cretinism, deaf-mutism, still births, ↓ IQ; **urinary iodine excretion** – best indicator of recent intake; **Global IDD Prevention Day – 21 October**; **ban on non-iodised salt for human consumption**
NPPCF (Prevention & Control of Fluorosis)	**2008–09**	**Fluoride > 1.5 mg/L** – dental mottling, skeletal fluorosis (**genu valgum**); **Nalgonda technique** – defluoridation (alum + lime); Gujarat, Rajasthan, Andhra affected
National Oral Health Programme	2014	Dental caries (fluoride < 0.5 mg/L), oral cancer screening
NPPMBI (Burn Injuries)	2010	Burn units
NPCCHH (Climate Change & Human Health)	2019	Heat-related illnesses, air pollution
National Programme for Palliative Care	2012	–

Programme	Year	Key points
National Tobacco Testing Labs, NPPC	–	–
National Sickle Cell Anaemia Elimination Mission	**1 July 2023 (Shahdol, MP)**	Screening of **7 crore people (0–40 yrs)** in tribal areas by 2025–26; **elimination by 2047**; sickle cell genetic status card
Anaemia Mukt Bharat	2018	**6 × 6 × 6 strategy** – 6 beneficiary groups, 6 interventions, 6 institutional mechanisms; **Test-Treat-Talk (T3) camps**
National Deworming Day	2015	**10 Feb & 10 Aug – albendazole 1–19 yrs**
Pradhan Mantri National Dialysis Programme	2016	Free dialysis at district hospitals

7.7 Important Health Days

Day	Date
World Leprosy Day / Anti-Leprosy Day (India)	**30 January**
World Cancer Day	**4 February**
World Hearing Day	**3 March**
National Vaccination Day	**16 March**
World Water Day	**22 March**
World TB Day	**24 March**
World Health Day	**7 April** (WHO foundation 1948)
World Malaria Day	**25 April**
World Red Cross Day	**8 May** (Henry Dunant's birthday)
International Nurses Day	**12 May** (Florence Nightingale)
National Dengue Day	**16 May**
World No Tobacco Day	**31 May**
World Environment Day	**5 June**
World Blood Donor Day	**14 June**
International Yoga Day	**21 June**
National Doctors' Day (India)	**1 July** (Dr B. C. Roy)
World Population Day	**11 July**
World Hepatitis Day	**28 July**
World Breastfeeding Week	**1–7 August**
National Nutrition Week	**1–7 September**
World Rabies Day	**28 September**
World Heart Day	**29 September**

Day	Date
International Day of Older Persons	**1 October**
World Mental Health Day	**10 October**
Global Hand Washing Day	**15 October**
World Polio Day	**24 October**
World Diabetes Day	**14 November** (Banting)
World AIDS Day	**1 December**
International Day of Persons with Disabilities	**3 December**
Universal Health Coverage Day	**12 December**

PYQ style

- NPCDCS launched – **2010**; renamed – **NP-NCD**.
- Population-based NCD screening age – **≥ 30 yrs**; checklist – **CBAC**.
- VIA uses – **3–5 % acetic acid**; age – **30–65 yrs, every 5 yrs**.
- Leading cause of blindness in India – **cataract**; programme – **NPCB 1976**.
- Blindness (WHO) – **VA < 3/60**.
- NMHP – **1982**; Mental Healthcare Act – **2017**; Tele-MANAS – **14416**.
- COTPA – **2003**; smoking in public places banned – **Section 4**; sale within **100 yards** of school banned.
- No sale of tobacco to – **< 18 yrs**; World No Tobacco Day – **31 May**.
- NPHCE – **2010**; elderly – **≥ 60 yrs**.
- Iodised salt at consumer level – **15 ppm**; NIDDCP – **1992**.
- Fluorosis – **> 1.5 mg/L**; defluoridation – **Nalgonda technique**.
- RPwD Act 2016 – **21 disabilities**; benchmark – **40 %**.
- World Health Day – **7 April**; International Nurses Day – **12 May**.

8 RMNCH+A, Family Welfare & Population Control

8.1 RMNCH+A Strategy (2013)

- **Reproductive, Maternal, Newborn, Child & Adolescent Health** – **"continuum of care"** across **life stages** and **levels of care (community → facility)**.
- **"+"** – adolescent health, and linking maternal & child health to reproductive health, family planning, life cycle approach.
- **High Priority Districts (HPDs)** – focused support.
- **5 × 5 matrix** – 5 high-impact interventions in 5 life stages.

Maternal health services (community nurse role)

- **Early registration (within 12 wks) & minimum 4 ANC visits** (1st – ≤ 12 wks, 2nd – 14–26 wks, 3rd – 28–34 wks, 4th – 36 wks–term); **MCP card**; **Td 2 doses**, **IFA 180 days**, **calcium 1 g**, **albendazole (2nd trimester)**, Hb, urine albumin/ sugar, BP, weight, **HIV, syphilis, HBsAg, blood group**, **GDM test (75 g OGTT)**.
- **Birth preparedness & complication readiness**; **institutional delivery**; **SBA (Skilled Birth Attendant)** – ANM/ LHV/ SN trained; **identify high-risk pregnancy (HRP)** – age < 18/ > 35, height < 145 cm, Hb < 7, BP ≥ 140/90, previous CS, multiple pregnancy, malpresentation, bleeding.
- **PNC** – **within 48 hrs, then day 3, 7, 42** (minimum); **HBNC visits by ASHA**.
- **Maternal Death Surveillance & Response (MDSR)**; **MAA, SUMAN, LaQshya**, **Midwifery led care units**.

Scheme	Key benefit
JSY (2005)	**Cash for institutional delivery** – LPS (rural ₹1,400, urban ₹1,000) & HPS (₹700/ ₹600); ASHA incentive
JSSK (2011)	**Free & cashless** delivery, CS, drugs, diagnostics, diet (3 days normal/ 7 days CS), blood, **transport (home–facility–home)** for pregnant women & sick infants
PMSMA (2016)	**Free ANC by specialists on 9th of every month** at PHCs/ CHCs
PMMVY (2017)	**₹5,000** for first child (2 instalments, **DBT**); **₹6,000 for second child if girl** (PMMVY 2.0, 2022)
SUMAN (2019)	Zero tolerance for preventable maternal/ newborn deaths – assured free services
LaQshya (2017)	Quality improvement in labour room & maternity OT

Child health services

- **Essential newborn care**, **HBNC**, **HBYC**, **immunization (UIP)**, **IMNCI**, **RBSK** (4 D's – mobile health teams screen 0–6 yrs at AWCs twice/ yr, 6–18 yrs in schools once/ yr), **SNCU/ NBSU/ NBCC**, **NRC (Nutrition Rehabilitation Centre)** for SAM, **IDCF/ STOP Diarrhoea campaign**, **SAANS**, **Vitamin A**, **Anaemia Mukt Bharat**, **National Deworming Day**.

- **Child Death Review (CDR)**.

Adolescent health – RKSK (2014)

- **10–19 yrs**; **6 strategic priorities** – nutrition, sexual & reproductive health, non-communicable diseases, substance misuse, injuries & violence (incl. GBV), mental health.
- **AFHC (Adolescent Friendly Health Clinics) – "Ujjwal" clinics**; **peer educators ("Saathiya")** – 4 per village (2 boys + 2 girls); **Saathiya Salah app/ resource kit**; **WIFS (weekly blue IFA tablet + biannual albendazole)**; **Menstrual Hygiene Scheme** (subsidised sanitary napkins by ASHA); **Adolescent Health Day (AHD)** quarterly.

8.2 Family Planning

- **Family planning (WHO)** – way of thinking & living adopted voluntarily to promote health & welfare of family.
- **History** – **India – first country in the world to launch a national family planning programme – 1952**; renamed **Family Welfare Programme (1977)**; **"cafeteria approach" → "Basket of choice"**; **target-free approach (1996)**; **Mission Parivar Vikas (2016)** – high-TFR districts.
- **Eligible couple**, **CPR**, **unmet need** (~9.4 % NFHS-5).

Method	Key points
Condom (Nirodh) – barrier	Free; also STI/ HIV protection; failure ~ 2–15 %
Combined OCP – Mala-N (free), Mala-D (social marketing)	21 hormonal + 7 iron tablets; **start day 1–5 of cycle**; inhibits ovulation
Centchroman – Chhaya (Saheli)	Non-steroidal, **twice weekly × 3 months then weekly**
Injectable DMPA – Antara	**150 mg IM every 3 months**
Emergency contraceptive pill (E-pill)	**LNG 1.5 mg within 72 hrs**
IUCD – Cu-T 380A (10 yrs), Cu 375 (5 yrs)	Insert within 12 days of menses; **PPIUCD (within 48 hrs of delivery)**, PAIUCD
Female sterilisation (tubectomy) – minilap/ laparoscopic	**Most popular method in India**; Pomeroy technique; postpartum within 7 days
Male sterilisation – NSV (No-Scalpel Vasectomy)	Simple, OPD, **backup contraception for 3 months (or 20 ejaculations/ semen analysis)**
Natural	**LAM** (exclusive BF + amenorrhoea + < 6 months), **Standard Days Method (CycleBeads – days 8–19)**, calendar, BBT, cervical mucus (Billings)

- **Spacing methods** – condoms, OCPs, IUCD, injectables, Chhaya; **terminal** – sterilisation.
- **Delivery of contraceptives by ASHA at doorstep** (Home Delivery of Contraceptives – 2011); **Family Planning Indemnity Scheme**; **compensation for sterilisation** (loss of wages); **Family Planning – Logistics Management Information System (FP-LMIS)**.
- **World Population Day – 11 July** (population fortnight: "Dampati Samparka Pakhwada" + "Seva Pakhwada").

8.3 Related Laws

Law	Key point
MTP Act 1971 (amended 2021)	**≤ 20 wks – 1 RMP**; **20–24 wks – 2 RMPs (special categories: rape survivors, minors, incest, disabled, foetal anomaly etc.)**; beyond 24 wks for **substantial foetal anomalies – Medical Board**; **contraceptive failure** ground extended to **unmarried women**; confidentiality
PCPNDT Act 1994 (amended 2003)	**Prohibits sex selection & disclosure of sex of foetus**; registration of USG machines; Form F; **Beti Bachao Beti Padhao (22 Jan 2015, Panipat)**
Prohibition of Child Marriage Act 2006	Girls 18, boys 21
Maternity Benefit (Amendment) Act 2017	**26 weeks paid maternity leave** (for first 2 children); crèche facility (≥ 50 employees)
Surrogacy (Regulation) Act 2021	Only altruistic surrogacy

PYQ style

- RMNCH+A launched – **2013**; concept – **continuum of care**.
- Minimum ANC visits – **4**; first ANC within – **12 weeks**.
- JSSK free diet for normal delivery – **3 days**; CS – **7 days**.
- PMSMA – **9th of every month**; PMMVY – **₹5,000**.
- India's family planning programme – **1952 (first in world)**.
- Most popular contraceptive method in India – **female sterilisation**.
- Antara – **DMPA 3-monthly**; Chhaya – **weekly non-steroidal**.
- Cu-T 380A – **10 yrs**; PPIUCD within – **48 hrs**.
- NSV backup contraception – **3 months**.
- MTP up to 20 wks – **1 RMP**; up to 24 wks (special) – **2 RMPs**.
- PCPNDT Act – **1994**; prohibits – **sex determination**.
- Maternity leave – **26 weeks**.
- World Population Day – **11 July**.
- RKSK peer educators – **Saathiya**; adolescent clinics – **AFHC (Ujjwal)**.

9 Nutrition in Community Health

9.1 Nutrients – Key Facts

Nutrient	Calories / Key points
Carbohydrate	**4 kcal/g**; 55–60 % of energy
Protein	**4 kcal/g**; **RDA adult ~0.83 g/kg/day** (ICMR 2020); **egg protein – reference protein (NPU ~ 96, biological value ~ 94)**; **cereal + pulse = mutual supplementation** (lysine in pulse, methionine in cereal)
Fat	**9 kcal/g**; essential fatty acids – linoleic, linolenic; **visible fat 20–30 % of energy**
Alcohol	7 kcal/g
Fibre	Prevents constipation, colon cancer

- **ICMR–NIN RDA 2020** – **reference man 65 kg, reference woman 55 kg**; adult **sedentary man ~2110 kcal/day, woman ~1660 kcal/day** (moderate man ~2710); **pregnancy + 350 kcal (2nd & 3rd trimester)**, **lactation + 600 (0–6 m)/ + 520 (6–12 m)**; protein pregnancy +9.5 g (2nd) / +22 g (3rd).
- **NIN (National Institute of Nutrition) – Hyderabad**; **"My Plate for the Day"** – ≥ half plate vegetables & fruits.
- **Balanced diet** – contains all nutrients in correct proportion for health; **food groups** – cereals, pulses, milk/ animal foods, vegetables & fruits, fats & sugars.

Mineral/ vitamin	Key points
Iron	RDA women 29 mg (2020); **heme iron (meat) better absorbed**; **vitamin C ↑ absorption; tea/ coffee, phytates, calcium ↓**
Calcium	RDA adult 1000 mg; milk best source; ragi rich
Iodine	150 µg/day adults; iodised salt
Vitamin A	Night blindness; green leafy vegetables, yellow fruits, liver
Vitamin D	Sunlight; rickets
Folic acid	NTD prevention
Vitamin C	Amla (richest natural), citrus; scurvy
Thiamine (B1)	Beriberi (polished rice); parboiling preserves
Niacin	Pellagra (maize/ jowar eaters – leucine excess)

9.2 Malnutrition – Community Burden & Assessment

- **Double burden** – undernutrition + overweight/ obesity.
- **NFHS-5 (2019–21) under-5** – **stunting 35.5 %**, **wasting 19.3 %**, **severe wasting 7.7 %**, **underweight 32.1 %**, **overweight 3.4 %**; **anaemia children 6–59 m – 67.1 %**, **women 57 %**.
- **Methods of nutritional assessment ("ABCD")** – **A**nthropometry (weight, height, MUAC, skinfold, BMI), **B**iochemical (Hb, serum albumin, urinary iodine), **C**linical examination (signs), **D**ietary survey

(24-hr recall, weighment, food frequency) + vital statistics, ecological factors.

- **Weight-for-age** – underweight; **height-for-age** – stunting (chronic); **weight-for-height** – wasting (acute); **MUAC < 11.5 cm** (6–59 m) – SAM.
- **BMI (Quetelet's index) = weight (kg) / height² (m²)**; **Asian cut-offs: normal 18.5–22.9, overweight 23–24.9, obese ≥ 25**; WHO international: overweight 25–29.9, obese ≥ 30. **Waist circumference** – Asian men > 90 cm, women > 80 cm (abdominal obesity).
- **Growth monitoring** – **MCP card growth chart** monthly (AWW) – **flattening/ falling curve** = action.

9.3 Nutrition Programmes

Programme	Year/ Ministry	Key points
ICDS (Integrated Child Development Services)	**2 Oct 1975**, MWCD	**Beneficiaries – children 0–6 yrs, pregnant & lactating mothers, adolescent girls (14–18, aspirational districts/ NE)**; **6 services** – **supplementary nutrition**, **immunization**, **health check-up**, **referral services**, **pre-school non-formal education (3–6 yrs)**, **nutrition & health education**; **Anganwadi worker + helper**; now under **Saksham Anganwadi & Poshan 2.0 (2021)**
Supplementary Nutrition (ICDS norms – Poshan 2.0)	–	**Children 6 m–6 yrs: 500 kcal + 12–15 g protein**; **severely malnourished: 800 kcal + 20–25 g protein**; **pregnant & lactating: 600 kcal + 18–20 g protein**; **Take Home Ration (THR)** for 6 m–3 yrs & P/L women; **Hot Cooked Meal (HCM)** 3–6 yrs
PM POSHAN (earlier Mid-Day Meal)	**1995** (renamed **2021**), Ministry of Education	Hot cooked meal to children in **Balvatika + classes 1–8** in govt/ aided schools; **primary: 450 kcal + 12 g protein**; **upper primary: 700 kcal + 20 g protein**
POSHAN Abhiyaan (National Nutrition Mission)	**8 March 2018 (Jhunjhunu, Rajasthan)**	Reduce **stunting, undernutrition, anaemia, LBW**; **convergence, ICT (Poshan Tracker app)**, **Jan Andolan**; **Poshan Maah – September**, **Poshan Pakhwada – March**
Anaemia Mukt Bharat	**2018**	**6 × 6 × 6**; IFA by age groups; T3 camps; deworming; fortified foods
NIDDCP	1992	Iodised salt
Vitamin A prophylaxis	1970	9 m – 5 yrs
Targeted PDS / PMGKAY	NFSA **2013**	**National Food Security Act** – **5 kg foodgrains per person/ month** to priority households; **Antyodaya Anna Yojana (AAY) – 35 kg per family**; **PMGKAY – free foodgrains (extended to 2028)**
Rice fortification	**2021–24**	**Fortified rice (iron, folic acid, B12)** through PDS, ICDS, PM POSHAN
Double fortified salt (DFS)	–	Iodine + iron
Kishori Shakti Yojana/ SABLA → Scheme for Adolescent Girls	–	Nutrition & life skills for adolescent girls

9.4 Food Hygiene, Fortification & Adulteration

- **Food fortification** – adding micronutrients to commonly consumed food (**iodised salt, vitamin A & D in edible oil & milk, fortified rice/ wheat flour**); **"+F" logo (FSSAI)**.
- **Food enrichment** – restoring nutrients lost in processing.
- **Food adulteration** – **Food Safety & Standards Act (FSSA) 2006** → **FSSAI (2008)** (replaced PFA Act 1954); **Eat Right India** campaign.

Adulterant	Food	Health effect
Kesari dal (Lathyrus sativus)	Arhar/ besan	**Lathyrism – spastic paraplegia (neurolathyrism) – BOAA toxin**
Argemone oil	Mustard oil	**Epidemic dropsy** (oedema, glaucoma)
Metanil yellow	Turmeric, sweets, dal	Carcinogenic, liver damage
Lead chromate	Turmeric	Anaemia, brain damage
Papaya seeds	Black pepper	Liver problems
Starch	Milk, paneer	–
Brick powder	Chilli powder	GI problems
Aflatoxin (Aspergillus flavus)	**Groundnut, maize (mouldy)**	**Hepatocellular carcinoma**, liver damage
Ergot (Claviceps purpurea)	Bajra, rye	Ergotism – gangrene

- **Milk hygiene** – **pasteurisation** (holder 63 °C × 30 min; HTST 72 °C × 15 sec); **phosphatase test – checks adequacy of pasteurisation**; **methylene blue reduction test – bacterial quality**; milk-borne diseases – TB, brucellosis, typhoid, Q fever.
- **Food poisoning** – **Staph aureus (1–6 hrs, preformed toxin)**, **Bacillus cereus (fried rice)**, **Clostridium botulinum (canned food – descending paralysis; most fatal)**, Clostridium perfringens, Salmonella (12–24 hrs).

PYQ style

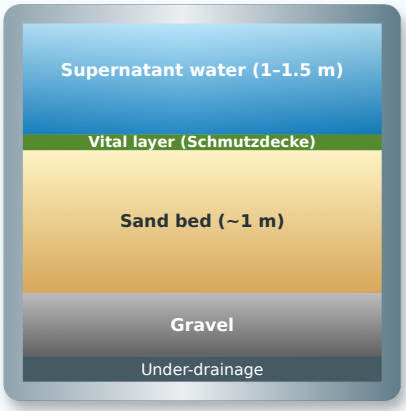
- Calories – carbohydrate/ protein **4 kcal/g**, fat **9 kcal/g**.
- Reference protein – **egg**.
- Extra energy in pregnancy – **+350 kcal**; lactation – **+600 kcal**.
- NIN located at – **Hyderabad**.
- Asian BMI obesity – **≥ 25**; overweight – **23–24.9**.
- ICDS started – **2 Oct 1975**; services – **6**.
- Supplementary nutrition for child (ICDS) – **500 kcal, 12–15 g protein**; pregnant – **600 kcal**.
- Mid-day meal renamed – **PM POSHAN (2021)**; primary – **450 kcal, 12 g protein**.
- POSHAN Abhiyaan – **8 March 2018, Jhunjhunu**; Poshan Maah – **September**.
- NFSA – **2013**; 5 kg grain per person.
- Lathyrism – **Kesari dal (BOAA)**; epidemic dropsy – **argemone oil**.
- Aflatoxin – **groundnut → liver cancer**.
- Pasteurisation check – **phosphatase test**.
- Most fatal food poisoning – **botulism**.
- Food safety authority – **FSSAI (FSS Act 2006)**.

10 Environmental Health & Sanitation

10.1 Water

- **Safe & wholesome water** – free from pathogens & harmful chemicals, pleasant to taste, usable for domestic purposes.
- **Requirement – 150–200 litres/ capita/ day (urban)**; **Jal Jeevan Mission (15 Aug 2019) – "Har Ghar Jal" – 55 litres/ person/ day functional household tap connection** in rural areas by 2024 (extended to 2028).
- **Sources** – rain (purest natural), surface (rivers, lakes, ponds – most polluted), **ground water (wells, springs – deep wells safest)**.
- **Water quality standards (BIS 10500)** – **pH 6.5–8.5**; **turbidity < 1 NTU (max 5)**; **fluoride 1.0 mg/L (permissible 1.5)**; **nitrate ≤ 45 mg/L (blue baby syndrome – methaemoglobinaemia)**; **arsenic ≤ 0.01 mg/L** (WB – skin lesions, cancer); **total hardness ≤ 200 mg/L**; **coliforms – 0 per 100 ml (E. coli – indicator of faecal contamination)**.
- **Hardness** – **temporary (bicarbonates – removed by boiling/ Clark's method – lime)**, **permanent (sulphates/ chlorides – removed by base exchange/ permutit process)**.

Water Purification – Slow Sand Filter, Chlorination & Well Disinfection



Slow Sand (Biological) Filter
Bacteria removal 99.9 %; rate 2–3 m³/m²/day

Chlorination – key numbers

Residual free chlorine:
0.5 mg/L after 1 hr contact

Chlorine demand → measured by
Horrock's apparatus

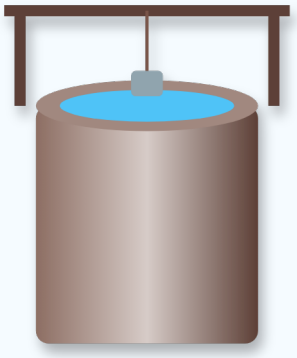
Residual chlorine test →
Orthotolidine (OT) test

Bleaching powder =
33 % available chlorine

Break-point chlorination –
beyond it, free chlorine appears

Chlorine tablet (Halazone):
for small quantities of water

Well disinfection



1. Find volume of water in well
2. Horrock's test → bleaching powder
3. Make paste → dissolve → pour
4. Contact 1 hr → OT test (0.5 mg/L)

Best done at night; well ≥ 15 m from latrine

Safe water: no coliform in 100 ml · fluoride ≤ 1 mg/L (max 1.5) · pH 6.5–8.5
Rapid sand filter: coagulation (alum) + frequent backwashing (urban)
Household: boiling (rolling boil 5–10 min) – best
Jal Jeevan Mission (2019): 55 litres/person/day tap water

Slow sand filter, chlorination facts & well disinfection

Purification of water

- **Large scale – storage** (sedimentation – 90 % suspended impurities settle; reduces bacteria ~90 % in 5–7 days; **Schistosoma cercariae die in 48 hrs**) → **filtration** → **chlorination**.

	Slow sand (biological/ English) filter	Rapid sand (mechanical/ American) filter
Rate	**2–3 m^3/m^2/day** (slow)	**80–200 m^3/m^2/day**

	Slow sand (biological/ English) filter	Rapid sand (mechanical/ American) filter
Pre-treatment	Plain sedimentation	**Coagulation (alum) + rapid mixing + flocculation**
Key layer	**Vital layer/ Schmutzdecke (zoogleal layer) – heart of filter**	–
Cleaning	**Scraping top layer** (weeks–months)	**Backwashing** (daily)
Bacterial removal	**99.9–99.99 %**	98–99 %
Space/ use	More space, small towns	Less space, **big cities**

- **Chlorination** – kills pathogens (not spores, cysts, some viruses); **chlorine demand** satisfied first; **free residual chlorine 0.5 mg/L (ppm) after 1 hour contact** (minimum; **0.2 mg/L at consumer end**); **break-point chlorination** – point where chlorine demand is fully met & free residual chlorine appears; **super-chlorination** (excess then dechlorination); **orthotolidine (OT) test** – residual chlorine (**yellow** colour; **OT-arsenite test** distinguishes free & combined chlorine).
- **Horrock's apparatus** – estimates **dose of bleaching powder** needed to disinfect water (6 white cups, 1 black cup, **starch-iodide indicator – blue colour**).
- **Bleaching powder (chlorinated lime) – 33 % available chlorine** (unstable – store in dark, dry, airtight container); **high-test hypochlorite (HTH) – 60–70 %**; **chlorine tablets** (0.5 g tablet for 20 L); **perchloron**.
- **Household purification** – **boiling (rolling boil 5–10 min – best, kills all pathogens incl. spores in some time)**, chlorination (tablets/ liquid), filtration (ceramic candles – **Pasteur–Chamberland, Berkefeld, Katadyn (silver)**), **SODIS** (solar), alum (fitkari) for turbidity, RO.
- **Well disinfection** – best method: **bleaching powder** (based on Horrock's test); steps – find volume (**$\pi r^2 h$; 1 m^3 = 1000 L**), dissolve bleaching powder in bucket (paste → milky solution), let lime settle, pour supernatant into well, **contact 1 hr**, **OT test (0.5 mg/L)**; best done **at night**; **sanitary well** – parapet wall, apron/ platform, drainage, covered, hand pump, **≥ 15 m (50 ft) from latrine/ source of contamination**.
- **Rain water harvesting**.

Water-borne & water-related diseases

- **Water-borne (faeco-oral)** – **cholera, typhoid, hepatitis A & E, polio, diarrhoea, dysentery**, giardiasis, amoebiasis.
- **Water-washed** (lack of water for hygiene) – scabies, trachoma, skin infections.
- **Water-based** (aquatic host) – **schistosomiasis, guinea worm (dracunculiasis – India eradicated 2000)**.
- **Water-related insect vector** – malaria, dengue, filaria.
- **Chemical** – **fluorosis, arsenicosis, nitrates (blue baby), lead**.

10.2 Air & Ventilation

- **Air pollutants** – **SO_2 (acid rain, respiratory)**, **CO (carboxyhaemoglobin – affinity ~200–250 × O_2)**, NO_2, ozone, **$PM_{2.5}$ & PM_{10} (most harmful – lungs)**, lead, hydrocarbons; **indoor air pollution** –

biomass fuel (chulha) – **COPD, pneumonia in children, LBW**; **PM Ujjwala Yojana (2016)** – LPG to BPL women.

- **AQI (Air Quality Index)** – **0–50 good**, 51–100 satisfactory, 101–200 moderate, 201–300 poor, 301–400 very poor, **401–500 severe**; **National Clean Air Programme (NCAP, 2019)**.
- **Greenhouse gases** – **CO_2 (main)**, methane, N_2O, CFCs → global warming; **ozone layer depletion – CFCs** (Montreal Protocol 1987).
- **Comfort indices** – **corrected effective temperature**, **Kata thermometer** (cooling power of air), **wet bulb globe temperature (heat stress)**; **comfortable zone in India 25–27 °C CET**.
- **Ventilation** – **air change 2–3 times/hr (living rooms)**; **cross ventilation**; **window area ≥ 1/5 floor area**; **floor space/ person ≥ 50–100 sq ft**.
- **Overcrowding** – > 2 persons per room / persons per room ratio; spreads TB, meningitis, scabies, ARI.

10.3 Excreta Disposal

- **Faecal-borne diseases** (cholera, typhoid, hepatitis, worms) – **"F diagram"** – **F**aeces → **F**ingers, **F**lies, **F**luids, **F**ields → **F**ood → new host; **sanitation barrier** blocks primary transmission.
- **Sanitary latrine** – **excreta should not contaminate surface/ ground water, not accessible to flies/ animals, no odour, simple, cheap, acceptable**.
- **Types (unsewered areas)** – **service type** (banned – **manual scavenging prohibited – Act 2013**), **bore-hole latrine**, **dug-well/ pit latrine**, **water-seal latrine (RCA type/ PRAI type)** – **water seal 2 cm (¾ inch)**, **twin-pit pour-flush latrine (Sulabh)** – most recommended for rural, **septic tank** (individual households – capacity **20–40 gallons (90–150 L)/ person**, retention ~1–3 days; anaerobic digestion; **sludge removed every 1–3 yrs**), **aqua privy**, **chemical closet** (aircraft, trains).
- **Sewered areas** – **water carriage system**; sewage treatment – **primary (screening, grit chamber, sedimentation)**, **secondary (biological – trickling filter, activated sludge process)**, **oxidation ponds/ stabilisation ponds (algae + bacteria – symbiosis)** – cheap for small towns.
- **BOD (Biochemical Oxygen Demand)** – **measure of organic pollution of sewage (5 days at 20 °C)**; domestic sewage ~ 300 mg/L; effluent standard ≤ 30 mg/L.
- **Swachh Bharat Mission – 2 Oct 2014**; **India declared ODF (Open Defecation Free) – 2 Oct 2019**; **SBM-G Phase II – ODF Plus** (solid & liquid waste management).

10.4 Solid Waste (Refuse) Disposal

- **Methods** – **composting** (**Bangalore method – anaerobic, hot fermentation**; **Indore method – aerobic, manual turning**; **mechanical composting**; **vermicomposting**), **sanitary landfill (controlled tipping)** – most satisfactory for cities (refuse compacted, covered with earth daily ~ 20–30 cm), **incineration** (hospital waste – best for infectious refuse), **dumping** (unsatisfactory), **manure pits** (rural), burial.
- **Solid Waste Management Rules 2016** – **segregation at source – green (wet/ biodegradable) & blue (dry)**, domestic hazardous.

10.5 Biomedical Waste Management Rules 2016 (amended 2018, 2019)

Bio-Medical Waste Management Rules 2016 – Colour Coding



YELLOW bag

- Human anatomical waste
- Animal waste
- Soiled waste (dressings, cotton, plaster, blood bags)
- Expired/ discarded medicines
- Chemical waste, lab cultures (pre-treated)

→ Incineration / plasma pyrolysis / deep burial



RED bag

Contaminated recyclable PLASTIC waste:

- IV sets, tubings, bottles
- Catheters, urine bags
- Syringes (without needle)
- Gloves, vacutainers

→ Autoclave/ microwave + shredding → recycle



WHITE (translucent)

Puncture-proof, leak-proof, tamper-proof:

- Needles, syringes with fixed needles
- Scalpel blades, lancets
- Any sharp metal

→ Autoclave + shredding/ encapsulation, sharp pit



BLUE box

Cardboard box with blue colour marking:

- Broken/ discarded glass (vials, ampoules, slides)
- Metallic body implants

→ Disinfection + recycling

Untreated BMW: store max 48 hrs · segregate at source · bar-code label · biohazard symbol

BMW Rules 2016 – four colour categories & treatment

Category	Container	Waste	Treatment
Yellow	**Non-chlorinated yellow plastic bag**	**Human anatomical**, animal, **soiled waste** (dressings, plaster casts, blood bags), **expired/ discarded medicines**, chemical waste, **chemical liquid**, discarded linen/ mattresses with blood, **microbiology lab & blood bank waste (pre-treated by autoclave/ NaOCl on site)**	**Incineration / plasma pyrolysis / deep burial** (rural, as per rules)
Red	**Non-chlorinated red plastic bag/ container**	**Contaminated recyclable waste** – tubing, bottles, **IV sets, catheters, urine bags, syringes (without needles)**, **gloves**, vacutainers	**Autoclave/ microwave/ hydroclave + shredding → recycling** (not incinerated)
White (translucent)	**Puncture-proof, leak-proof, tamper-proof container**	**Sharps** – needles, syringes with fixed needles, needles from needle tip cutter, **scalpels, blades**, metal sharps	Autoclave/ dry heat sterilisation + **shredding/ mutilation/ encapsulation** in metal container/ sharp pit
Blue	**Cardboard box with blue marking** (puncture-proof)	**Glassware – broken/ discarded/ contaminated glass incl. vials, ampoules**; **metallic body implants**	Disinfection (soak in **1 % NaOCl**)/ autoclave → **recycling**

- **Key rules** – **segregation at source**; **untreated BMW not stored > 48 hrs**; **bar-code labelling** of bags/ containers; **pre-treatment of lab/ microbiology waste on-site**; **phase out of chlorinated plastic bags** (prevents dioxin on burning); **biohazard symbol** (and cytotoxic symbol for cytotoxic waste – yellow bag with "cytotoxic" label); **annual report to SPCB** by 30 June; **training & immunization (Hep B) of all health workers**; **records for 5 yrs**; **CBWTF (Common Bio-Medical Waste Treatment Facility)** – within **75 km**.

- **Liquid waste** – disinfected before discharge into drains.
- **Needle-stick injury** – wash with soap & water (don't squeeze), report, PEP evaluation.
- **Incinerator** – **double chamber; primary 800 ± 50 °C, secondary 1050 ± 50 °C**.

10.6 Housing, Noise, Radiation & Entomology

- **Healthy housing criteria (WHO)** – **floor area per person ≥ 50 sq ft (min) – 100 sq ft ideal**, **room height ≥ 3 m (10 ft)**, **window area 1/5 of floor**, **lighting – daylight factor ≥ 1 %**, separate kitchen, safe water, sanitary latrine.
- **Noise** – unit **decibel (dB)**; **normal conversation ~60 dB**; **> 85 dB (8 hrs) – hearing loss (occupational)**; **> 120 dB – pain threshold**; effects – **auditory (temporary/ permanent threshold shift, noise-induced hearing loss – 4000 Hz notch)**, non-auditory (sleep, BP, stress); **Noise Pollution Rules 2000**; residential limit **day 55 dB, night 45 dB**; **silence zone (hospitals, schools) – day 50 dB, night 40 dB** (within **100 m**).
- **Radiation** – **ionising** (X-rays, γ-rays) – cancer, genetic effects; **ALARA principle**; **lead apron**; **radiation unit – sievert (Sv)**; **TLD badge/ film badge** for workers; **maximum permissible dose – 20 mSv/yr (5-yr average) occupational**; **Atomic Energy Regulatory Board (AERB)**. **Non-ionising** – UV (cataract, skin cancer), microwaves.
- **Medical entomology** –

Arthropod	Diseases
Anopheles	Malaria
Aedes	Dengue, chikungunya, Zika, yellow fever
Culex	**Filaria (bancroftian), JE**, West Nile
Mansonia	**Brugian filaria** (Pistia plant)
Housefly	Typhoid, cholera, diarrhoea, trachoma (mechanical)
Sandfly	**Kala-azar**, oriental sore, sandfly fever
Rat flea (Xenopsylla cheopis)	**Plague**, murine typhus; **cheopis index > 1 – plague risk**
Louse	**Epidemic typhus**, relapsing fever, trench fever
Tick	**KFD (Kyasanur Forest Disease)**, Lyme, tick typhus
Trombiculid mite (chigger)	**Scrub typhus**
Itch mite	Scabies
Cyclops	Guinea worm, fish tapeworm
Tsetse fly	Sleeping sickness (Africa)

- **Insecticides** – **DDT** (organochlorine – banned for agriculture; restricted vector control), **malathion**, **temephos (Abate – larvicide in potable water)**, **synthetic pyrethroids (deltamethrin, cyfluthrin – IRS, LLINs)**, **pyrethrum (space spray/ fogging)**; **IGR**; biological – **Gambusia, Guppy fish, BTI**.

PYQ style

- Jal Jeevan Mission – **55 L/person/day**, launched **2019**.
- Heart of slow sand filter – **vital layer (Schmutzdecke)**; cleaning – **scraping**.
- Rapid sand filter cleaning – **backwashing**; coagulant – **alum**.
- Residual chlorine – **0.5 mg/L after 1 hr contact**.
- Chlorine demand – **Horrock's apparatus**; residual chlorine – **orthotolidine test**.
- Bleaching powder – **33 % available chlorine**.
- Best household purification – **boiling**.
- Indicator of faecal contamination – **E. coli / coliforms**; standard – **0 per 100 ml**.
- Blue baby syndrome – **nitrates**.
- Temporary hardness removed by – **boiling**.
- Latrine–well distance – **≥ 15 m (50 ft)**.
- Water seal in latrine – **2 cm**.
- ODF India – **2 Oct 2019**; SBM – **2 Oct 2014**.
- BOD measures – **organic pollution of sewage**.
- Best method for city refuse – **sanitary landfill**; hospital waste – **incineration**.
- BMW Rules – **2016**; untreated storage max – **48 hrs**.
- Anatomical waste – **yellow**; IV sets/ catheters – **red**; needles – **white (translucent)**; broken glass – **blue**.
- Hearing loss – **> 85 dB**; silence zone around hospital – **100 m**.
- Plague vector – **rat flea**; KFD – **tick**; scrub typhus – **mite**.

11 Occupational Health

11.1 Basics

- **Occupational health (ILO/ WHO, 1950)** – promotion & maintenance of the **highest degree of physical, mental & social well-being of workers in all occupations**; **"adaptation of work to man and of each man to his job"** (ergonomics).
- **Father of occupational medicine – Bernardino Ramazzini** (**De Morbis Artificum** – 1700).
- **Ergonomics** – fitting the job to the worker.

Hazard type	Examples
Physical	Heat (heat stroke, cramps), cold, light, **noise (NIHL)**, vibration (**Raynaud's – "white fingers"**), radiation, pressure (**caisson disease – decompression sickness**)
Chemical	Dusts, gases, metals (lead, mercury, chromium), solvents (benzene – leukaemia)
Biological	**Brucellosis, anthrax (wool sorter's disease), leptospirosis, Q fever**, TB, tetanus, hepatitis B/ HIV (health workers)
Mechanical	Accidents, injuries
Psychosocial	Stress, shift work, burnout
Ergonomic	Back pain, repetitive strain injury, carpal tunnel

11.2 Pneumoconiosis & Lung Diseases

Disease	Dust	Occupation / key point
Silicosis	**Free silica (SiO_2)**	**Mining, quarrying, stone cutting, sand blasting, pottery, glass**; **most common & most serious pneumoconiosis**; **↑ TB risk (silicotuberculosis)**; **"snowstorm" X-ray, eggshell calcification of hilar nodes**; irreversible; **notifiable**
Anthracosis	**Coal dust**	**Coal miners** ("black lung")
Asbestosis	**Asbestos fibres**	Mining, insulation, brake lining; **mesothelioma (pleura), lung cancer**; **asbestos bodies**
Byssinosis	**Cotton fibre dust**	**Textile mills**; **"Monday fever/ Monday chest tightness"**
Bagassosis	**Sugarcane fibre (bagasse)**	Paper/ cardboard industry; **thermophilic actinomycetes**
Farmer's lung	**Mouldy hay/ grain**	**Thermophilic actinomycetes (Micropolyspora faeni)** – hypersensitivity pneumonitis
Siderosis	Iron oxide	Welders, iron workers (benign)
Berylliosis	Beryllium	Electronics
Tobacosis	Tobacco dust	Tobacco industry

- **Dust particles 0.5–3 µm** (respirable) reach alveoli – most dangerous.

11.3 Occupational Poisonings & Cancers

Agent	Key points
Lead (plumbism)	**Most common occupational metal poisoning**; battery, paint, printing, smelting; **inhalation – main route**; **Burtonian blue line (gums)**, **abdominal colic**, constipation, anaemia (**basophilic stippling**), wrist drop, encephalopathy; **↑ coproporphyrin in urine (CPU), ↑ ALA, blood lead**; Rx – **chelation ($CaNa_2EDTA$, DMSA, BAL)**
Mercury	Tremors, erethism, gingivitis, acrodynia; **Minamata disease (methyl mercury – fish)**
Chromium	**Nasal septum perforation**, chrome ulcers, lung cancer
Arsenic	Raindrop pigmentation, keratosis, skin cancer
Benzene	**Leukaemia, aplastic anaemia**
Carbon monoxide	Headache, **cherry-red colour**; hyperbaric O_2
Methyl isocyanate (MIC)	**Bhopal gas tragedy (2–3 Dec 1984)** – Union Carbide; **National Pollution Control Day – 2 Dec**
Organophosphates	Farmers – cholinergic crisis; **atropine + PAM**

Occupational cancer	Agent
Skin cancer	**Soot, coal tar, mineral oil** (scrotal cancer in chimney sweeps – **Percival Pott, 1775 – first occupational cancer**), arsenic, UV
Lung cancer	Asbestos, chromium, nickel, arsenic, radon (uranium miners)
Bladder cancer	**Aromatic amines (aniline dyes – benzidine, β-naphthylamine)** – dye & rubber industry
Leukaemia	**Benzene**, ionising radiation
Liver (angiosarcoma)	**Vinyl chloride**
Mesothelioma	**Asbestos**
Nasal sinus	Wood dust, nickel

- **Occupational dermatoses** – **most common occupational disease overall** (contact dermatitis – cement, chromium, rubber, chemicals).
- **Occupational hazards of health workers** – **needle-stick injuries (HBV most transmissible – 30 %, HCV 3 %, HIV 0.3 %)**, TB, back injuries, radiation, chemicals (glutaraldehyde, cytotoxic), violence, stress.
- **Agricultural workers** – pesticides, snake bite, zoonoses (leptospirosis, brucellosis), heat stroke, farmer's lung, injuries.

11.4 Prevention

- **Medical measures** – **pre-placement (pre-employment) examination** (baseline, fit for job), **periodic medical examination** (early detection), medical & health care services, notification, supervision of working environment, **records**, health education, **first aid**.
- **Engineering measures (most effective)** – **substitution** (non-toxic for toxic – e.g. sandblasting with non-silica), **enclosure/ isolation**, **local exhaust ventilation**, **wet methods (dust suppression)**,

mechanisation/ automation, design of building, good housekeeping, environmental monitoring (**TLV – threshold limit values**).

- **Personal protective equipment (PPE)** – masks/ respirators, ear plugs/ muffs, goggles, gloves, aprons, helmets, safety shoes – **last line of defence**.
- **Legislation** – see below.
- **Hierarchy of controls** – **elimination > substitution > engineering > administrative > PPE**.

11.5 Labour Laws & Social Security

Act	Key points
Factories Act 1948 (amended 1987)	Applies to factories with **≥ 10 workers (with power) / ≥ 20 (without power)**; **minimum age 14 yrs**; **working hours ≤ 48 hrs/week, ≤ 9 hrs/day**; adolescents (15–18) ≤ 4.5 hrs/day; **no woman at night (7 PM – 6 AM)** except with safeguards; **canteen (≥ 250 workers)**, **crèche (≥ 30 women workers)**, **first-aid box (1 per 150 workers)**, **ambulance room (≥ 500 workers)**, **welfare officer (≥ 500)**, **safety officer (≥ 1000)**; annual leave with wages (1 day per 20 days worked); **notifiable diseases to Chief Inspector of Factories**
ESI Act 1948 (Employees' State Insurance)	**First implemented – 24 Feb 1952 (Kanpur & Delhi)**; covers **non-seasonal factories/ establishments with ≥ 10 employees**, **wage ceiling ₹21,000/month (₹25,000 for disabled)**; contribution – **employer 3.25 %, employee 0.75 %** (employees earning < ₹176/day exempt from contribution); administered by **ESIC** (Ministry of Labour)
ESI benefits (7)	**Medical benefit** (full, for insured person & family), **sickness benefit (70 % of wages, up to 91 days/yr)**, **maternity benefit (100 % wages, 26 weeks)**, **disablement benefit** (temporary – 90 % of wages; permanent – as per loss of earning capacity), **dependants' benefit** (on death from employment injury – 90 % of wages shared), **funeral expenses (₹15,000)**, **unemployment allowance (Rajiv Gandhi Shramik Kalyan Yojana/ ABVKY)**; + rehabilitation, confinement expenses
Employees' Compensation Act 1923 (Workmen's Compensation)	Compensation for injury/ occupational disease/ death arising out of employment
Maternity Benefit Act 1961 (amended 2017)	26 weeks leave; crèche (≥ 50 employees)
Mines Act 1952, **Plantations Labour Act 1951**, **Dock Workers Act**	Sector-specific
Child & Adolescent Labour Act 1986 (amended 2016)	No child < 14 yrs in any occupation; 14–18 not in hazardous
New Labour Codes (2019–20) – **Code on Social Security 2020**, **Occupational Safety, Health & Working Conditions Code 2020**	Subsume older Acts (implemented in phases)
ILO	**Geneva**, est. **1919**

- **Institutes** – **National Institute of Occupational Health (NIOH) – Ahmedabad, Gujarat** (ICMR); **Central Labour Institute – Mumbai**.

- **Sickness absenteeism** – index of health of workers.

PYQ style

- Father of occupational medicine – **Ramazzini**.
- Most common & serious pneumoconiosis – **silicosis**; predisposes to – **TB**.
- Byssinosis – **cotton dust (Monday fever)**; bagassosis – **sugarcane**; anthracosis – **coal**.
- Farmer's lung – **mouldy hay (thermophilic actinomycetes)**.
- Mesothelioma – **asbestos**.
- Most common occupational metal poisoning – **lead**; sign – **Burtonian line, basophilic stippling**.
- Bladder cancer – **aniline dyes**; scrotal cancer in chimney sweeps – **soot (Percival Pott)**.
- Vinyl chloride – **angiosarcoma of liver**; benzene – **leukaemia**.
- Bhopal gas – **methyl isocyanate (1984)**.
- Most effective preventive measure – **engineering (substitution, enclosure, ventilation)**; PPE – **last line**.
- ESI Act – **1948**; wage ceiling – **₹21,000**; employer contribution – **3.25 %**.
- ESI sickness benefit – **70 % wages, 91 days**.
- Factories Act – crèche when **≥ 30 women**; canteen **≥ 250 workers**; working hours **48/week**.
- NIOH located at – **Ahmedabad**.

12 Health Education, Communication & IEC/ BCC

12.1 Health Education

D Health Education (National Conference on Preventive & Social Medicine, USA)

"A process that **informs, motivates and helps people to adopt & maintain healthy practices and lifestyles**, advocates environmental changes as needed to facilitate this goal and conducts professional training & research to the same end." Health education aims at **changing knowledge → attitude → practice/ behaviour (KAP)**.

- **Aims/ objectives (WHO)** – encourage people to **adopt & sustain health-promoting lifestyles**; make **judicious use of health services**; take own decisions to improve health status (**individual & collective responsibility – self-reliance**).
- **Approaches** – **regulatory (legal)**, **service**, **health education** (most effective in long term), **primary health care** approach.
- **Models** – **medical model**, **motivation model**, **social intervention model**.
- **Contents** – human biology, nutrition, hygiene, family health, disease prevention & control, mental health, prevention of accidents, use of health services.

Principles of health education

1. **Credibility** – message consistent with scientific knowledge & local culture.
2. **Interest** – people won't learn what they're not interested in (**felt needs**).
3. **Participation** – **active participation – high level of learning** (principle of "**learning by doing**").
4. **Motivation** – awaken desire; **primary** (inborn – hunger, sex) & **secondary** (rewards, praise, love); **incentives**.
5. **Comprehension** – **level of understanding**, local language, avoid jargon.
6. **Reinforcement** – repetition of message at intervals.
7. **Learning by doing** – Chinese proverb "**If I hear, I forget; if I see, I remember; if I do, I know**".
8. **Known to unknown** – start where people are.
9. **Setting an example** – health worker should practise what is taught.
10. **Good human relations**.
11. **Feedback**.
12. **Leaders** – use local leaders (opinion leaders, teachers, religious leaders) as agents of change.

- **Stages of adoption of new idea (Rogers' diffusion of innovation)** – **Awareness → Interest → Evaluation → Trial → Adoption**; adopter categories – innovators (2.5 %), early adopters, early majority, late majority, laggards.
- **Behaviour change models** – **Health Belief Model** (perceived susceptibility, severity, benefits, barriers, cues to action, self-efficacy), **Stages of change (Transtheoretical – Prochaska & DiClemente)**: **pre-contemplation → contemplation → preparation → action → maintenance** (± relapse), PRECEDE-PROCEED, social learning theory (Bandura).

12.2 Methods of Health Communication

Category	Methods	Notes
Individual approach	**Personal contact, home visits, counselling, personal letters, telephone**	**Most effective** for behaviour change; time-consuming
Group approach	**Lecture/ chalk-talk**, **demonstration**, **group discussion**, **panel discussion**, **symposium**, **workshop**, **seminar**, **conference**, **role play**, **brainstorming**, **Philips 66 (buzz session)**, **case study**, **film shows**, **puppet show**, **exhibition**	Two-way, cost-effective
Mass approach	**Television, radio, newspapers, posters, health magazines, internet/ social media, mobile (SMS), cinema, folk media** (street plays – nukkad natak, bhavai in Gujarat), **hoardings**	Reaches large numbers; **one-way – limited behaviour change**; creates awareness

Group methods – details

Method	Description
Lecture	One-way; **group ≤ 30**; **duration 15–20 min**; supplement with AV aids
Demonstration	**"Show how" – carefully prepared presentation** – e.g. **ORS preparation, hand washing, breastfeeding** – **"seeing is believing"**; **result demonstration** (shows outcome) vs **method demonstration** (shows how)
Group discussion	**6–12 members**, leader + recorder; circular seating; **each one participates**; free exchange
Panel discussion	**4–8 qualified persons (experts) discuss a topic in front of a large audience**; chairman; **audience participates at end**
Symposium	**Series of speeches by experts on a selected subject**; **no discussion among speakers**; audience may ask questions at end
Workshop	**Series of meetings (4 or more)**, emphasis on **individual work within group (problem solving)**, 10–25 participants
Seminar	Group of experts/ learners study a topic intensively under guidance
Conference	Large meeting to share knowledge; many sessions
Role play	Participants **enact a real situation** – learn attitudes, communication
Brainstorming	**Free spontaneous ideas** from group, no criticism; then evaluated
Philips 66 / buzz session	Large group divided into groups of **6, discuss for 6 min**

12.3 Audio-Visual (AV) Aids

Type	Examples
Auditory	Radio, tape recorder, public address system, podcasts
Visual – projected	**Slides, filmstrips, overhead projector (OHP), LCD/ PowerPoint, epidiascope**

Type	Examples
Visual – non-projected	**Chalkboard (blackboard – most commonly used)**, **flannel graph**, **flash cards** (series of 10–12 cards, talk with each), **flip charts**, **posters (one idea, few words, bold)**, charts, **models, specimens, exhibits**, pamphlets, leaflets
Combined (audio-visual)	**Television, films/ videos, cinema**, demonstrations, puppetry

- **Poster** – **attract attention, single message, simple, large letters**, placed at eye level, changed frequently.
- **Edgar Dale's "Cone of Experience"** – from **most concrete (base) – direct purposeful experiences** → contrived experiences (models) → dramatised → demonstrations → field trips → exhibits → TV/ films → recordings, radio, pictures → visual symbols → **verbal symbols (most abstract – top)**. **Learning retention** – people remember ~10 % of what they read, 20 % hear, 30 % see, 50 % see & hear, 70 % say, **90 % say & do**.

12.4 Communication

- **Communication** – process of **sharing information, ideas, feelings** to bring about **understanding**; Latin communis (common).
- **Elements** – **Sender (communicator) → Message → Channel (medium) → Receiver (audience) → Feedback** (+ effects); **SMCR model (Berlo)**.
- **Types** – **one-way (didactic – lecture)** vs **two-way (Socratic – discussion)**; **verbal** (oral/ written) vs **non-verbal** (body language, gestures, facial expression, eye contact – conveys ~ 2/3 of message); formal vs informal (**grapevine**); **upward, downward, horizontal**; **visual**, **telecommunication**.
- **Barriers** – **physiological** (deafness, poor vision, aphasia), **psychological** (emotional disturbance, neurosis, prejudice, fear), **environmental** (noise, invisibility, crowding), **cultural** (illiteracy, language, beliefs, customs, attitudes – **most important in India**), semantic (jargon), status differences.
- **Key communication skills** – **active listening**, empathy, clarity, feedback, respect, culturally appropriate.
- **IEC** – Information, Education & Communication; **BCC** – Behaviour Change Communication (targeted to specific audience to change behaviour); **SBCC** – Social & BCC; **IPC** – interpersonal communication (most effective for behaviour change).
- **Health communication strategies** – **advocacy** (to policy makers), **social mobilisation** (community, partners), **BCC** (individuals).

12.5 Counselling

- **Counselling** – **two-way, face-to-face** process to **help a person make decisions & solve problems** (helping relationship); based on **trust, confidentiality, empathy, non-judgemental attitude**, client-centred.
- **GATHER approach (family planning counselling)** – **G**reet, **A**sk, **T**ell, **H**elp, **E**xplain, **R**eturn visit.
- **Pre-test & post-test counselling (HIV)** – informed consent, confidentiality (**ICTC**).
- **Steps** – establish rapport, explore problem, identify options, help decision, plan action, follow-up.
- **Counselling vs health education** – counselling individual, decision-oriented; education informs.

12.6 Planning a Health Education Programme

1. **Assess needs** (felt needs, community diagnosis) → 2. **Set objectives (SMART)** → 3. **Identify target group & content** → 4. **Select methods & media** → 5. **Pre-test materials** → 6. **Implement** → 7. **Evaluate (process, impact, outcome)**.

PYQ style

- Health education aims to change – **KAP (knowledge, attitude, practice)**.
- Most effective approach for behaviour change – **individual (personal contact/ counselling)**.
- Principle "learning by doing" – **participation**.
- Stages of adoption – **awareness → interest → evaluation → trial → adoption**.
- Group discussion size – **6–12**; lecture – **≤ 30, 15–20 min**.
- Experts discuss in front of audience – **panel discussion (4–8)**.
- Series of speeches, no discussion among speakers – **symposium**.
- Demonstration best for – **skills (ORS preparation, hand washing)**.
- Most commonly used visual aid – **chalkboard**.
- Most concrete in Edgar Dale's cone – **direct purposeful experience**; most abstract – **verbal symbols**.
- Communication elements – **sender, message, channel, receiver, feedback**.
- Most important barrier in India – **cultural (illiteracy, beliefs)**.
- FP counselling – **GATHER approach**.
- Stages of change – **pre-contemplation → contemplation → preparation → action → maintenance**.

13 Home Visit, Family Health Care, Records & Reports

13.1 Family Health Nursing

- **Family – basic unit of society & of community health nursing service**. Types – **nuclear**, **joint (extended)**, **three-generation**, single-parent, blended.
- **Functions of family** – reproductive, **socialisation**, economic, affectional, **health care**, protection, status.
- **Family health nursing process** – **assessment (family health assessment – family folder/ family health card)** → **nursing diagnosis (family health problems)** → **planning with family** → implementation → evaluation.
- **Family Health Tree/ genogram**, **eco-map**.
- **Family life cycle (Duvall – 8 stages)** – beginning family → childbearing → preschool → school-age → teenage → launching → middle-aged parents → ageing family.

13.2 Community Diagnosis & Survey

- **Community diagnosis** – identification & quantification of health problems of a community, based on data (morbidity, mortality, environment, resources) – **basis of community health planning**.
- **Community survey** steps – define objectives, area, prepare tools (**schedule/ questionnaire**), pre-test, train surveyors, **map** the area, collect data (house-to-house), analyse, report, plan.
- **Community needs assessment** – **felt needs** (people's perceived) vs **real needs** (professionally identified); **community participation** essential.
- **PRA (Participatory Rural Appraisal)** – **social mapping, seasonal calendar, transect walk, Venn diagram**.
- **Windshield survey** – quick observation driving/ walking through community.

13.3 Home Visit

D Home visit

A **family-nurse contact** that allows the health worker to **assess the home & family situation** to provide necessary nursing care & health-related activities. **Backbone/ cornerstone of community health nursing**.

- **Purposes** – give care to sick, **health education in family setting**, observe home environment, **case finding & follow-up**, **contact tracing**, referral, MCH services (ANC/ PNC, newborn), build rapport, demonstrations.
- **Principles** – **purposeful (definite objective)**, **planned in advance**, flexible, **based on family needs**, **family participation**, **regular & continuous**, use of resources, **records maintained**, health education integral, **courtesy & respect for customs**, punctuality, confidentiality.
- **Steps** – **(1) Initiation** (contact, appointment), **(2) Pre-visit preparation** (review records, plan, prepare bag), **(3) In-home activities** (introduce, greet, explain purpose, **establish rapport**, assess, **care using**

bag technique, health education, plan next visit), **(4) Termination** (summarise, plan follow-up), **(5) Post-visit – recording & reporting**, evaluation.

- **Priority of home visits** – **acute communicable disease/ sick > newborn & postnatal mother > antenatal > infant > under-5 > others**; visit **non-infectious cases first and infectious cases last** in a day (to prevent spread).



CHN/ public health bag – steps of bag technique

Bag technique

- **Bag technique** – tool by which the nurse, during home visit, **performs nursing procedures with ease & deftness, saving time & effort, without contaminating the bag & its contents**; **purpose – prevent cross-infection** (nurse ↔ patient ↔ family ↔ community).
- **Principles** – bag to have **all necessary articles**; **place bag on a clean, flat, high surface** (never on bed/ floor), **on paper lining**; **hand washing before & after care**; **open bag only after washing hands**; **separate clean & dirty articles**; **apron worn with clean side inside & folded with soiled side in**; take out only what is needed; **clean & disinfect reusable articles before putting back**; bag **cleaned regularly & contents replenished**.
- **Contents (typical)** – **thermometers (oral & rectal)**, **BP apparatus, stethoscope**, **scissors (2)**, **forceps**, **kidney tray**, syringes & needles, **gauze, cotton**, **spirit, savlon/ antiseptic**, **soap in soap dish, hand towel, apron**, **newspapers**, **gloves**, mackintosh, **urine testing kit (Benedict's/ dipsticks)**, test tubes, spirit lamp, **ORS, IFA, paracetamol, contraceptives**, **MCP cards, records, IEC material**, measuring tape, **weighing scale** (separately).

13.4 Clinics in Community

- **MCH clinic, ANC clinic, under-5 clinic, immunization session (VHND – Village Health & Nutrition Day; now VHSND)**, family planning, NCD clinics, TB/ leprosy clinics.
- **VHSND (Village Health, Sanitation & Nutrition Day)** – **monthly at Anganwadi**; ANM, ASHA, AWW – immunization, ANC, growth monitoring, health education, contraceptives, referral.
- **Under-5 clinic (Morley's concept)** – care in illness, **immunization**, **growth monitoring (Road to Health chart)**, oral rehydration, family planning, health education; **"Road to Health card"** – symbol:

triangle of care.

13.5 Records & Reports

- **Record** – **written account of an event/ activity** (permanent, legal document); **report** – summary of records/ activities **communicated to others** (higher authority) in a prescribed form.
- **Importance** – continuity of care, **legal document**, communication, evaluation, **planning & policy**, research, statistics, supervision, teaching.
- **Principles** – **accurate, complete, concise, legible, timely (written promptly), signed & dated**, **confidential**, in ink/ digital, no erasures (single line cross & initial), standardised forms.

Registers/ records at Sub-Centre (ANM)	Purpose
Eligible Couple Register (ECR)	Couples with wife 15–49 yrs; **updated every April (annual)**; FP methods
MCH register / RCH register	Pregnant women, deliveries, PNC, child immunization
Birth & death register	Vital events
Immunization/ due list	Tracking
Village/ family health register (family folder)	Household survey
Stock register (drugs, vaccines, equipment)	Inventory
Communicable disease registers (fever – malaria, TB, leprosy)	Surveillance
NCD register (CBAC)	Screening
MCP card (with mother)	Home-based record

- **Digital systems** – **HMIS (Health Management Information System, 2008)**, **RCH portal (earlier MCTS – Mother & Child Tracking System, 2009)** – name-based tracking with **RCH ID**, **Ni-kshay**, **IHIP**, **ANMOL app (ANM Online)**, **U-WIN** (digital immunization registry, 2023–24), **eVIN**, **ABHA**, **TeCHO+ (Gujarat – Technology Enabled Community Health Operations)**.
- **Monthly reports** – ANM → PHC (by 5th of month approx.) → block → district; **weekly IDSP S-form**.

13.6 Referral System

- **Referral** – transferring a patient to a higher/ specialised level for better management; **two-way (referral + feedback/ back-referral)**.
- **Referral slip** – patient details, findings, treatment given, reason, contact; **ASHA escorts**; **108 ambulance (emergency)**, **102 (pregnant women & sick infants – JSSK)**.

13.7 Roles of ANM/ Female Health Worker & CHO (Job responsibilities)

- **ANM (FHW)** – **MCH** (ANC, delivery, PNC, newborn care, HBNC support), **immunization**, **family planning** (IUCD insertion, OCP, condoms), **nutrition** (IFA, vitamin A, ORS, zinc), **communicable disease control** (fever surveillance, RDT, DOT, leprosy), **NCD screening**, **health education**, **records & reports** (registers, RCH portal), **VHSND**, supervise/ support **ASHA & dais**, **school health**, coordinate with AWW, **first aid**.

- **MPW (male)** – vector control, malaria (RDT/ smear), sanitation, chlorination of wells, TB/ leprosy, environmental sanitation, health education to men, NSV motivation.
- **LHV (Health Supervisor – female)** – **supervision & guidance of ANMs** (~4–6 SCs), conducts clinics, checks records, training, conducts difficult deliveries, liaison with MO.
- **CHO (Community Health Officer, AAM-SC)** – **team leader**; delivers **CPHC 12 packages**, NCD screening & follow-up, **teleconsultation (eSanjeevani)**, dispensing essential drugs, wellness sessions, **performance-linked incentives**; supervises ASHAs & MPWs at SC level.

PYQ style

- Basic unit of CHN service – **family**.
- Backbone of community health nursing – **home visit**.
- First step of home visit – **initiation / pre-visit preparation (planning)**.
- In a day, infectious cases visited – **last**.
- Purpose of bag technique – **prevent cross-infection**.
- CHN bag placed on – **clean, flat, high surface with newspaper**.
- Apron – **clean side inside when folded (soiled side in)**.
- VHSND held – **monthly at Anganwadi Centre**.
- Under-5 clinic concept – **David Morley (Road to Health chart)**.
- Eligible couple register updated – **every April**.
- Record vs report – record **permanent account**; report **summary sent to authority**.
- Name-based tracking of mothers & children – **RCH portal (earlier MCTS)**.
- Gujarat's community health app – **TeCHO+**.
- LHV supervises – **ANMs**.
- Ambulance for pregnant women & sick infants – **102**; emergency – **108**.

14 Health of Special Groups – School, Urban, Tribal, Elderly, Disabled & Others

14.1 School Health

- **First school medical inspection in India – Baroda (Vadodara), Gujarat – 1909**.
- **School Health Committee (Renuka Ray) – 1960**; **school health services** recommended by Bhore Committee.
- **Objectives** – promote positive health, prevent disease, early diagnosis & treatment, **health-conscious citizens**, healthy school environment.
- **Aspects of school health services** – **health appraisal** (medical examination at entry & periodically – every 4 yrs (older), teachers' observation daily), **remedial measures & follow-up**, **prevention of communicable diseases (immunization – Td at 10 & 16 yrs)**, **healthful school environment**, **nutritional services (PM POSHAN)**, first aid & **emergency care**, **mental health**, **dental health**, **eye health (vision screening – refractive errors – most common problem detected)**, **health education**, **education of handicapped children**, **school health records** (cumulative).
- **School environment standards** – **location** away from noise/ traffic, **site 10 acres (higher secondary)**, **classroom – 10 sq ft per student (floor space)**, **≤ 40 students per class**, **desks – "minus" type (desk overhangs seat)**, **window area ≥ 25 % (1/4) of floor area**, lighting from left side, **drinking water**, **separate latrines (1 urinal per 60 students, 1 latrine per 100 students)**.
- **Common health problems in school children** – **malnutrition, anaemia, infections (worms, skin – scabies), dental caries, refractive errors, ear infections**, behaviour problems, learning disorders.
- **Ayushman Bharat – School Health & Wellness Programme (Feb 2020, MoHFW + MoE)** – **2 teachers per school as "Health & Wellness Ambassadors"**; **11 thematic areas** (growing up healthy, emotional well-being, interpersonal relationships, values & citizenship, gender equality, nutrition & hygiene, substance misuse, healthy lifestyle, reproductive health & HIV, violence & injury, internet safety); **Tuesdays as Health & Wellness Day**; **RBSK screening** for 6–18 yrs.
- **Kayakalp, Swachh Vidyalaya**.

14.2 Urban Health

- **Urbanisation** – India ~**31 % urban (Census 2011)**; **slums – 17 % of urban population** (Census 2011); problems – overcrowding, poor sanitation, **air & water pollution**, vector-borne diseases (dengue, malaria – **An. stephensi**), TB, RTI, NCDs, substance abuse, violence, migrants.
- **NUHM (1 May 2013)** – **UPHC per 50,000** (slum/ vulnerable), **UCHC per 2.5–5 lakh**, **urban ASHA (per 1,000–2,500 slum population)**, **Mahila Arogya Samiti (MAS – 50–100 households)**, **outreach sessions (UHND)**, **Urban AAM (Ayushman Arogya Mandir – urban) per 15,000–20,000**.
- **Smart Cities Mission, AMRUT, PMAY (housing)**.

14.3 Tribal Health

- **Tribal population ~8.6 % (Census 2011)**; Gujarat ~14.8 % (Dangs district majority tribal).
- **Problems** – **malnutrition, anaemia, sickle cell disease (high in Gujarat tribal belts)**, malaria, TB, skin & worm infections, **high IMR/ MMR**, poor access, superstitions, alcohol, geographic isolation.

- **Special norms – SC per 3,000, PHC per 20,000, CHC per 80,000**; **mobile medical units**, **birth waiting homes**, **ASHA per habitation**, **Eklavya schools**, **Van Bandhu Kalyan Yojana (Gujarat, 2007)**, **National Sickle Cell Anaemia Elimination Mission (2023)**, **PM-JANMAN (2023) for PVTGs**.

14.4 Geriatric (Elderly) Health

- **Elderly – ≥ 60 yrs** (India); **~8.6 % (Census 2011) → ~10.5 % (2022) of population** (projected ~20 % by 2050); **"feminisation of ageing"**.
- **Problems** – **NCDs (HTN, DM, CVD, arthritis – osteoarthritis, osteoporosis)**, **cataract, hearing loss**, **dementia (Alzheimer's)**, depression, **falls & fractures (hip)**, incontinence, **polypharmacy**, malnutrition, **loneliness, neglect & elder abuse**, financial dependency.
- **NPHCE (2010)**; **Maintenance & Welfare of Parents & Senior Citizens Act 2007** (maintenance tribunals); **National Policy on Older Persons 1999**; **National Council for Senior Citizens**; **Indira Gandhi National Old Age Pension Scheme (IGNOAPS)**; **Rashtriya Vayoshri Yojana** (aids & assistive devices); **Atal Vayo Abhyuday Yojana**; **Elder Line 14567**; **Ayushman Vay Vandana – PMJAY for all ≥ 70 yrs (2024)**; **International Day of Older Persons – 1 Oct**.
- **Nursing** – comprehensive geriatric assessment (**ADL – Katz index, IADL – Lawton**), fall prevention (lighting, grab bars, non-slip mats), nutrition (calcium, vit D, fibre, fluids), immunization (influenza, pneumococcal), medication review, social support, day-care centres, palliative care.

14.5 Disability & Rehabilitation

- **Disability** – WHO ICF; ~2.2 % (Census 2011), ~**15 % globally (WHO)**.
- **RPwD Act 2016** – **21 types**, benchmark ≥ 40 %, **UDID card**, reservation (jobs 4 %, education 5 %), **Accessible India Campaign (Sugamya Bharat Abhiyan, 2015)**.
- **Community-Based Rehabilitation (CBR)** – WHO strategy (1978) – rehabilitation, equalisation of opportunities & social inclusion **using community resources** – **CBR matrix: health, education, livelihood, social, empowerment**.
- **Rehabilitation types** – medical, vocational, social, psychological; **District Disability Rehabilitation Centres (DDRC)**, **ADIP scheme** (aids & appliances).
- **Mental health in community** – **DMHP**, early identification, reduce stigma, **community care (half-way homes, day care)**, ASHA/ ANM training (**mhGAP**), suicide prevention.

14.6 Other Groups

Group	Key concerns
Women	Anaemia, maternal health, violence (**One Stop Centre "Sakhi", Women helpline 181**), **Beti Bachao Beti Padhao (2015)**, PM Ujjwala, nutrition, menstrual hygiene
Adolescents	RKSK, nutrition, mental health, substance use, early marriage/ pregnancy
Industrial workers	Occupational health, ESI
Migrants & construction workers	TB, HIV, poor access – **e-Shram portal**, mobile clinics
Prisoners	TB, HIV, mental illness, substance abuse
Transgender persons	**Transgender Persons (Protection of Rights) Act 2019**; inclusive services

Group	Key concerns
Homeless	Shelters under NULM

PYQ style

- First school medical inspection in India – **Baroda, 1909**.
- School Health Committee – **Renuka Ray, 1960**.
- Most common defect in school children screening – **refractive error / dental caries** (also malnutrition, anaemia).
- Floor space per student – **10 sq ft**; students per class – **≤ 40**.
- Desk type – **minus desk**; window area – **¼ of floor area**.
- School Health & Wellness Ambassadors – **2 teachers per school**.
- UPHC population – **50,000**; Mahila Arogya Samiti – **NUHM**.
- Tribal SC norm – **3,000**.
- Elderly in India – **≥ 60 yrs**; NPHCE – **2010**.
- Senior citizens Act – **2007**; Elder Line – **14567**.
- ADL assessment – **Katz index**.
- CBR uses – **community resources**; CBR matrix – **5 components**.
- Women helpline – **181**; One Stop Centre – **Sakhi**.

15 Disaster Management & Emergency Services

15.1 Disaster – Basics

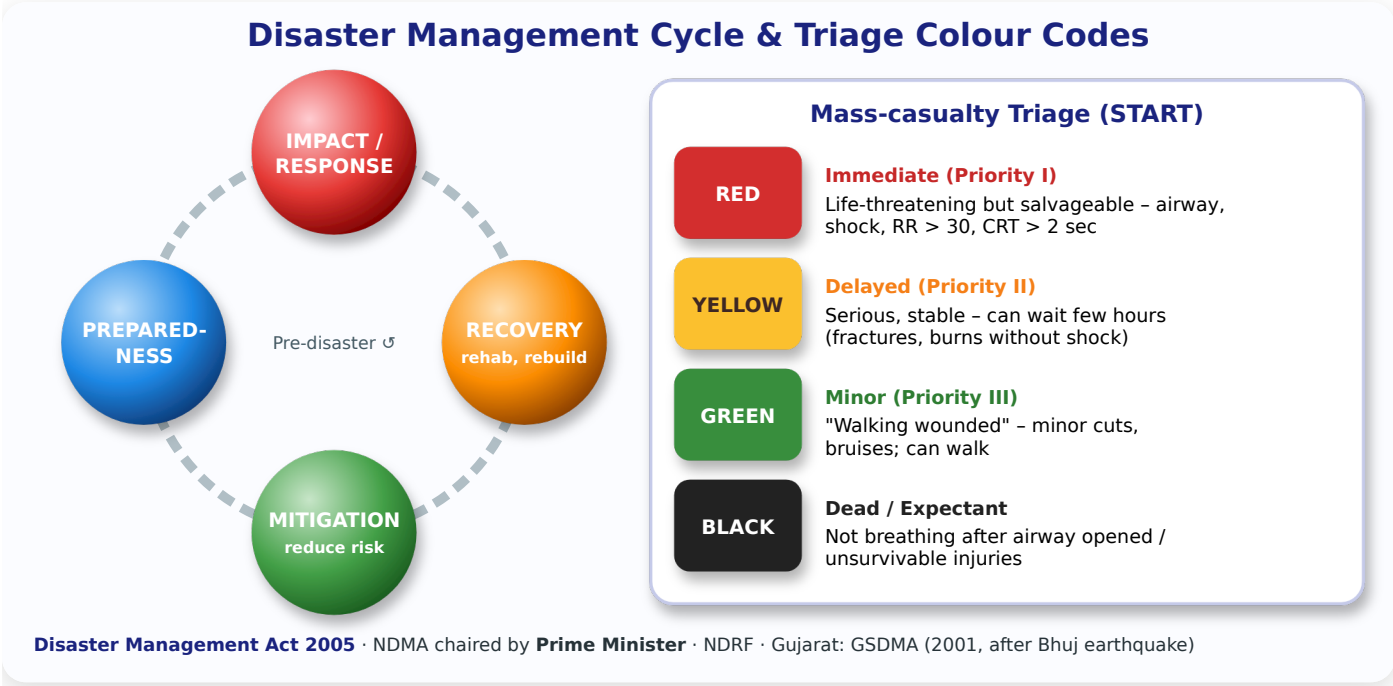
D Disaster (DM Act 2005)

A **catastrophe, mishap, calamity or grave occurrence** arising from natural or man-made causes, resulting in **substantial loss of life, human suffering, damage to property/ environment**, beyond the **coping capacity of the affected community**. **Risk = Hazard × Vulnerability / Capacity**.

Natural	Man-made
Earthquake (Bhuj, Gujarat – **26 Jan 2001**, ~20,000 deaths), **cyclone** (Gujarat coast – Tauktae 2021, Biparjoy 2023), **flood**, **drought**, landslide, tsunami (**26 Dec 2004**), heat wave, cold wave, avalanche, lightning, epidemics/ pandemics	**Industrial/ chemical (Bhopal 1984)**, **nuclear (Chernobyl 1986, Fukushima 2011)**, fire, building collapse (**Morbi bridge collapse, Gujarat – 30 Oct 2022**), road/ rail/ air accidents, **stampede**, war, terrorism, bioterrorism

- **India** – ~**58.6 % landmass prone to earthquakes**, ~12 % floods, ~68 % drought, **5,700 km coast – cyclones**. **Gujarat** – earthquake (Kutch – **Zone V – highest seismic zone**), cyclones, floods, droughts, industrial hazards.

15.2 Disaster Management Cycle



Disaster management cycle & START triage colour codes

Phase	Activities
Mitigation (prevention)	Reduce risk/ impact – **structural** (earthquake-resistant buildings, embankments, cyclone shelters) & **non-structural** (land-use planning, building codes, insurance, awareness)
Preparedness	**Disaster plans**, **early warning systems**, **mock drills**, stockpiling, **training**, community awareness, hospital disaster plan, resource inventory, communication network
Response (impact/ emergency)	**Rescue (search & rescue)**, **evacuation**, **triage**, first aid, **mass casualty management**, shelter, food, water, sanitation, **disease surveillance**, identification & disposal of dead
Recovery (rehabilitation & reconstruction)	**Rehabilitation** (restoration of services, psychosocial support, livelihood) & **reconstruction** ("**build back better**"); long-term development

- **Pre-disaster** – mitigation & preparedness; **post-disaster** – response & recovery.
- **Sendai Framework for DRR (2015–2030)** – 4 priorities, 7 targets; replaced **Hyogo Framework (2005–15)**.

15.3 Disaster Management Structure in India

- **Disaster Management Act – 2005** (enacted 23 Dec 2005).
- **NDMA (National Disaster Management Authority)** – **chaired by Prime Minister**; policies, plans, guidelines.
- **NEC (National Executive Committee)** – chaired by **Union Home Secretary**; **Ministry of Home Affairs – nodal ministry** (drought – Agriculture; epidemics/ biological – **MoHFW**).
- **NDRF (National Disaster Response Force)** – specialised force (~16 battalions); **NIDM (National Institute of Disaster Management)** – training & research (New Delhi).
- **SDMA** – chaired by **Chief Minister**; **GSDMA (Gujarat State Disaster Management Authority) – 2001 (after Bhuj earthquake)**; **Gujarat State Disaster Management Act 2003** (first state DM Act); **SDRF**.
- **DDMA** – chaired by **District Collector/ DM**; **EOC (Emergency Operation Centre)** at each level.
- **National Policy on DM – 2009**; **National DM Plan – 2016** (first; updated 2019).
- **Disaster helpline – 1078/ 1070**; **International Day for Disaster Risk Reduction – 13 Oct**.

15.4 Mass Casualty Management & Triage

- **Triage** – sorting casualties by **priority of treatment** based on severity & likelihood of survival – **"to do the greatest good for the greatest number"**.
- **START (Simple Triage & Rapid Treatment)** – based on **RPM: Respiration (> 30/min), Perfusion (CRT > 2 sec / no radial pulse), Mental status (can't follow simple commands)**; **walking wounded → green**; **JumpSTART** for children.

Colour	Priority	Examples
Red	**Immediate (I)** – life-threatening but salvageable	Airway obstruction, tension pneumothorax, shock, severe haemorrhage, RR > 30
Yellow	**Delayed/ Urgent (II)** – serious but stable, can wait	Fractures (closed), moderate burns without shock, abdominal injuries stable
Green	**Minor (III)** – "walking wounded"	Minor cuts, abrasions, sprains

Colour	Priority	Examples
Black	**Dead / expectant (IV)** – unsalvageable	No breathing after airway opening, massive head injury

- **Field care** – **search & rescue**, **triage**, **first aid**, **stabilisation**, **evacuation** (transport to appropriate facility), **tagging (triage tags)**; **"golden hour"** (first hour after injury).
- **Hospital disaster plan** – **Incident Command System (ICS)**, surge capacity (discharge stable patients, cancel elective surgeries), triage area, communication, **mock drills**.

15.5 Public Health Problems after Disasters

- **Water-borne** – **diarrhoea, cholera, typhoid, hepatitis A & E** (floods – contaminated water); **vector-borne** – malaria, dengue (stagnant water – after 2–3 weeks); **leptospirosis (floods – rat urine)**; **ARI, measles** (overcrowding in camps – **measles vaccination priority in camps**, vitamin A); **malnutrition**; **injuries, tetanus (earthquakes)**; **snake bites (floods)**; **mental health – PTSD, depression, anxiety, grief**; NCDs interrupted care; **heat stroke/ hypothermia**.
- **Relief camp priorities** – **safe water (minimum 15 L/person/day – Sphere standards)**, **sanitation (1 latrine per 20 persons)**, shelter (**3.5 m^2 covered area/ person**), food (**2100 kcal/ person/ day**), immunization (measles), **disease surveillance & early warning (IDSP)**, vector control, health education, MCH care, **psychological first aid**.
- **Management of dead bodies** – **dead bodies don't cause epidemics** (unless cholera/ etc.); identification, dignified disposal; mass graves avoided.
- **Water disinfection in floods** – chlorination of wells, **chlorine tablets**, boiling.

15.6 Nurse's Role in Disaster

- **Preparedness** – participate in planning, **training (first aid, triage, CPR)**, mock drills, community awareness, maintain emergency supplies, vulnerable population list (pregnant, elderly, disabled).
- **Response** – **triage**, first aid, emergency care, **infection control**, **care of vulnerable groups**, immunization, nutrition, **disease surveillance & reporting**, sanitation supervision in camps, **psychosocial support (psychological first aid – "Look, Listen, Link")**, documentation.
- **Recovery** – rehabilitation, follow-up, community mental health, home visits, rebuild health services.
- **Self-care** of responders (burnout, compassion fatigue).

15.7 Emergency Medical Services

- **108 – Emergency Response Service (EMRI – GVK)** – **started in Gujarat – Aug 2007** (also AP 2005); free; **free ambulance for medical, police & fire emergencies**; **Basic Life Support (BLS) & Advanced Life Support (ALS) ambulances**.
- **102** – **JSSK free transport** for pregnant women & sick infants ("**Khilkhilat**" in Gujarat – drop back home for mother & newborn).
- **104** – health helpline (information, counselling); **1097** – AIDS helpline; **1098** – CHILDLINE; **181** – women helpline; **112** – national emergency number (Emergency Response Support System).
- **First aid principles** – **ABC**, **DR ABC** (Danger, Response, Airway, Breathing, Circulation), stop bleeding (**direct pressure**), immobilise fractures, prevent shock, treat burns (cool water 20 min), **don't give anything by mouth to unconscious**, arrange transport.

PYQ style

- DM Act – **2005**; NDMA chairman – **Prime Minister**; SDMA – **Chief Minister**; DDMA – **Collector**.
- Nodal ministry for DM – **Home Affairs**; biological disasters – **Health**.
- GSDMA established – **2001 (after Bhuj earthquake)**.
- Bhuj earthquake – **26 January 2001**.
- Disaster cycle – **mitigation, preparedness, response, recovery**.
- Triage principle – **greatest good for greatest number**.
- Red tag – **immediate**; yellow – **delayed**; green – **walking wounded**; black – **dead/ expectant**.
- START triage parameters – **respiration, perfusion, mental status**.
- Common disease after floods – **diarrhoea, leptospirosis**; in camps – **measles (vaccinate)**.
- Sphere water standard – **15 L/person/day**; latrine – **1 per 20**.
- 108 started in Gujarat – **2007**; 102 – **pregnant women & sick infants**.
- Sendai Framework – **2015–2030**.

16 Health Planning, Management, International Agencies & Health Insurance

16.1 Health Planning

- **Planning** – deciding in advance what to do, how, when & by whom; **planning cycle** – **analysis of health situation → establishment of objectives & goals → assessment of resources → fixing priorities → write-up of formulated plan & programmes → programming & implementation → monitoring → evaluation** (→ back to start).
- **Goal** (ultimate desired state, not time-bound), **objective** (specific, measurable, time-bound – **SMART**: Specific, Measurable, Achievable, Relevant/ Realistic, Time-bound), **target** (quantified), **activities**.
- **Monitoring** – day-to-day follow-up of activities during implementation; **evaluation** – assessing **relevance, adequacy, accessibility, acceptability, effectiveness, efficiency, impact** (done at end/ periodically).
- **Planning Commission (1950)** → replaced by **NITI Aayog (1 Jan 2015)** – chaired by PM; **Five-Year Plans (1951–2017; 12th last)**; now **15-year vision, 7-year strategy, 3-year action agenda**.

Management techniques

Technique	Use
PERT (Program Evaluation & Review Technique)	Network analysis – time estimation (optimistic, pessimistic, most likely)
CPM (Critical Path Method)	Longest path – minimum project duration
Cost-benefit analysis	Benefits in money terms
Cost-effectiveness analysis	Cost per unit outcome (e.g. per life saved)
Systems analysis	Choose best among alternatives
Management by Objectives (MBO) – Peter Drucker	Joint goal setting & review
SWOT analysis	Strengths, Weaknesses, Opportunities, Threats
Input–output analysis, **network analysis**, **Gantt chart**	Scheduling
Work sampling, Delphi technique (expert consensus)	–

- **Functions of management (POSDCORB – Luther Gulick)** – **P**lanning, **O**rganising, **S**taffing, **D**irecting, **CO**ordinating, **R**eporting, **B**udgeting.
- **Supervision** – guiding, helping & teaching staff to do their best (**supportive supervision**); tools – checklists, **supervisory visits**, records review, feedback; **span of control**.
- **Leadership styles** – **autocratic** (crisis), **democratic/ participative** (best for health teams), **laissez-faire** (free rein), situational.

- **Delegation**, **decision making**, **conflict management**, **time management**, **budgeting**, **inventory control** (**ABC analysis – value; VED – vital, essential, desirable**).
- **Indian Public Health Standards (IPHS)** – first 2007, revised **2012 & 2022**; **NQAS (National Quality Assurance Standards)**, **Kayakalp (2015 – cleanliness awards)**, **LaQshya**, **MusQan** (child-friendly).

16.2 International Health Agencies

Agency	Est./ HQ	Key points
WHO (World Health Organization)	**7 April 1948, Geneva** (World Health Day)	**UN specialised agency for health**; **World Health Assembly** (supreme body, meets annually in May), **Executive Board**, **Secretariat (Director-General – Dr Tedros, 2017)**; **6 regions** – **SEARO HQ – New Delhi** (India's region); functions – international health regulations, epidemiological surveillance, standards (essential medicines list, ICD), research, technical cooperation; **India – member since 1948**
UNICEF (UN Children's Fund)	**1946, New York**	Originally UN International Children's Emergency Fund; **child survival – GOBI-FFF** (**G**rowth monitoring, **O**ral rehydration, **B**reastfeeding, **I**mmunization, **F**amily spacing, **F**emale education, **F**ood supplementation); **Nobel Peace Prize 1965**; cold chain support
UNFPA	1969, New York	Population & reproductive health
UNDP	1965, New York	Development; **HDI**
FAO (Food & Agriculture Organization)	1945, **Rome**	Food security, nutrition
ILO (International Labour Organization)	**1919, Geneva**	Workers' health & rights
World Bank (IBRD)	1944, Washington DC	Loans for health projects
UNAIDS	1996, Geneva	HIV/ AIDS
UNESCO	1945, Paris	Education, science, culture
WFP	1961, Rome	Food aid
International Red Cross (ICRC)	**1863, Geneva – Henry Dunant**	Emblem – red cross on white; **World Red Cross Day 8 May** (Dunant's birthday); **Indian Red Cross Society – 1920** (President – President of India; Chairman – Union Health Minister); Geneva Conventions
Gavi (Vaccine Alliance)	2000, Geneva	Vaccines (Pentavalent, PCV, rotavirus support)
Global Fund	2002, Geneva	AIDS, TB, malaria
CDC (USA), **USAID**, **DANIDA (Denmark)**, **SIDA (Sweden)**, **NORAD**, **CIDA (Canada)**, **JICA (Japan)**, **DFID/ FCDO (UK)**	Bilateral	–

Agency	Est./ HQ	Key points
Rockefeller Foundation (1913), **Ford Foundation**, **Bill & Melinda Gates Foundation**	Philanthropic	–
CARE	1945	Food & nutrition
Colombo Plan	1950	Technical cooperation (Asia–Pacific)

- **Voluntary/ national agencies** – **Indian Red Cross Society (1920)**, **Bharat Sevak Samaj**, **Kasturba Memorial Fund**, **Hind Kusht Nivaran Sangh (leprosy, 1950)**, **Tuberculosis Association of India (1939 – TB seal)**, **All India Blind Relief Society**, **Family Planning Association of India (FPAI, 1949)**, **Indian Council for Child Welfare**, **Central Social Welfare Board (1953)**, **St John Ambulance**, **TNAI (Trained Nurses' Association of India, 1908)**, **Indian Nursing Council (1947 Act; 1949)**, **SEWA (Ahmedabad – self-employed women)**.

16.3 Health Insurance & Financing

- **Health financing in India** – govt health expenditure ~**1.9–2.1 % of GDP**; **out-of-pocket expenditure (OOPE)** high (~**39–40 %** of total health expenditure, declining).
- **Health insurance schemes**:

Scheme	Key points
CGHS (Central Government Health Scheme)	**1954, Delhi**; central govt employees & pensioners
ESIS (ESI Scheme)	**1948 Act** – industrial workers (see occupational health)
AB-PMJAY	**23 Sept 2018** – **₹5 lakh per family per year**, secondary & tertiary hospitalisation, **cashless & portable**; **all ≥ 70 yrs covered (2024)**; ~1,900+ packages; **National Health Authority**
RSBY (Rashtriya Swasthya Bima Yojana)	2008 – ₹30,000 for BPL (merged into PMJAY)
PMSBY (Pradhan Mantri Suraksha Bima Yojana)	Accident insurance ₹2 lakh @ **₹20/yr**
PMJJBY (Jeevan Jyoti)	Life insurance ₹2 lakh @ ₹436/yr
Community-based health insurance	**SEWA (Ahmedabad), Yeshasvini (Karnataka)**
Private insurance	IRDAI regulated

- **Universal Health Coverage (UHC)** – all people receive needed health services without financial hardship; **UHC Day – 12 Dec**.

16.4 Gujarat-Specific Health Schemes & Facts (GSSSB focus)

Scheme/ Fact	Key points
Chiranjeevi Yojana (2005)	**Free institutional delivery by private obstetricians** for BPL women (PPP) – ↑ institutional deliveries; won Wall Street Journal's Asian Innovation Award (2006)
Bal Sakha Yojana (2009)	Newborn care of BPL babies by **private paediatricians**
Mukhyamantri Amrutum (MA) Yojana (2012)	Tertiary care for BPL; merged with PMJAY → **PMJAY-MA – ₹10 lakh per family**
Khilkhilat (2012)	Free **drop-back van** for mother & newborn from hospital to home
108 Emergency Services (2007)	GVK-EMRI; free
Kasturba Poshan Sahay Yojana	Cash assistance to pregnant women (BPL) for nutrition
Doodh Sanjivani Yojana	**Fortified flavoured milk** to school children (tribal areas) & pregnant/ lactating women
Poshan Sudha Yojana	Hot cooked meal to pregnant & lactating women at Anganwadi (tribal)
Mamta Divas	Weekly (Wednesday) **immunization & MCH day** at village level – Gujarat's VHND
Mamta Taruni (adolescent girls)	Adolescent health services
Shala Arogya – RBSK	School health check-ups (Gujarat's school health programme)
TeCHO+	Digital community health platform (tablet-based) for ANM/ FHW
e-Mamta	Name-based mother & child tracking (Gujarat – before national MCTS)
NIOH	National Institute of Occupational Health – **Ahmedabad**
GSDMA	2001; **Gujarat State Disaster Management Act 2003**
Indian Institute of Public Health, Gandhinagar (IIPH-G)	Public health training (PHFI)

PYQ style

- Planning cycle first step – **analysis of health situation**.
- SMART objectives – **Specific, Measurable, Achievable, Relevant, Time-bound**.
- POSDCORB – **Luther Gulick** (management functions).
- Network technique for time estimation – **PERT**; longest path – **CPM**.
- Inventory control by value – **ABC**; by criticality – **VED**.
- NITI Aayog replaced Planning Commission – **1 Jan 2015**.
- WHO established – **7 April 1948, Geneva**; SEARO – **New Delhi**.
- UNICEF – **1946, New York**; GOBI-FFF – **UNICEF**.
- FAO HQ – **Rome**; ILO – **1919, Geneva**.
- Red Cross founder – **Henry Dunant (1863)**; Indian Red Cross – **1920**.
- CGHS started – **1954, Delhi**.
- Chiranjeevi Yojana – **Gujarat, private obstetricians, BPL deliveries (2005)**.
- Khilkhilat – **drop-back service for mother & newborn**.
- Gujarat PMJAY-MA cover – **₹10 lakh**.
- Mamta Divas – **Gujarat's village MCH & immunization day**.

17 Quick Revision – Master Tables & 150 One-Liners

17.1 Master Table – Numbers & Norms

Topic	Value
Sub-centre (plain / hilly-tribal)	**5,000 / 3,000**
PHC	**30,000 / 20,000**
CHC	**1,20,000 / 80,000** – 30 beds, 4 specialists
UPHC	**50,000**
ASHA	**1 per 1,000**, age **25–45 yrs**
Anganwadi centre	**400–800 population**
Residual chlorine	**0.5 mg/L after 1 hr**
Bleaching powder	**33 % available chlorine**
Fluoride (BIS)	**1.0 mg/L (max 1.5)**
Coliforms	**0 per 100 ml**
Water (JJM)	**55 L/person/day**
Latrine–well distance	**≥ 15 m**
Iodised salt (production / consumer)	**30 / 15 ppm**
Noise causing hearing loss	**> 85 dB**
Silence zone	**100 m** around hospitals/ schools
BMW storage (untreated)	**≤ 48 hrs**
Autoclave	**121 °C, 15 lb/in², 15–20 min**
Hot air oven	**160 °C × 2 hrs**
Flash pasteurisation	**72 °C × 15–20 sec**
Holder pasteurisation	**63 °C × 30 min**
Normal curve ±1 / 2 / 3 SD	**68.3 / 95.4 / 99.7 %**
Replacement fertility	**TFR 2.1 / NRR 1**
India TFR (NFHS-5)	**2.0**
IMR / NMR / U5MR (SRS 2021)	**27 / 19 / 31**
MMR (2019–21)	**93 per lakh live births**
Census 2011 sex ratio / child sex ratio	**943 / 919**
Literacy (2011)	**74.04 %**
Registration of birth/ death	**21 days**
School classroom	**10 sq ft per student; ≤ 40 students**

Topic	Value
Relief camp water / latrine	**15 L/person/day; 1 per 20**
ESI wage ceiling	**₹21,000/month**
ESI contribution (employer / employee)	**3.25 % / 0.75 %**
PM-JAY	**₹5 lakh/family/yr**; Gujarat PMJAY-MA **₹10 lakh**
Ni-kshay Poshan	**₹1,000/month**
TB regimen	**2HRZE + 4HRE**
Leprosy MDT	**PB 6 months, MB 12 months**
Leprosy elimination	**< 1/10,000**
PEP for HIV	**within 72 hrs (best 2 hrs), 28 days**
Rabies ID schedule	**0, 3, 7, 28 (0.1 ml × 2 sites)**
Yellow fever quarantine	**6 days**

17.2 Master Table – Years & Persons

Event	Year
Bhore Committee	**1946**
Mudaliar Committee	**1962**
Chadah Committee	**1963**
Mukerji Committee	**1965–66**
Jungalwalla Committee	**1967**
Kartar Singh Committee	**1973**
Shrivastava Committee	**1975**
ICDS	**1975**
Alma-Ata	**1978**
NHP	**1983, 2002, 2017**
Ottawa Charter	**1986**
NRHM / NUHM	**2005 / 2013**
RMNCH+A	**2013**
Ayushman Bharat (HWC / PMJAY)	**2018**
WHO	**1948**
UNICEF	**1946**
ILO	**1919**
Red Cross / Indian Red Cross	**1863 / 1920**
India FP programme	**1952**
NACO	**1992**

Event	Year
RNTCP / NTEP	1997 / 2020
NMHP / DMHP	1982 / 1996
NPCB	1976
NPCDCS / NP-NCD	2010 / 2023
NPHCE	2010
COTPA / NTCP	2003 / 2007
IDSP	2004
BMW Rules	2016
DM Act	2005
ESI Act / Factories Act	1948 / 1948
Swachh Bharat / ODF India	2014 / 2019
POSHAN Abhiyaan	2018
Smallpox eradicated / India	1980 / 1977
Polio-free India	2014
Father of epidemiology (modern)	John Snow
Father of public health nursing	Lillian Wald
Father of occupational medicine	Ramazzini
Father of demography	John Graunt

17.3 150 One-Liners

One-liners 1–50

1. WHO health definition – **1948**.
2. Most sensitive health indicator – **IMR**.
3. PQLI components – **IMR, life expectancy at 1, literacy**.
4. DALY – **YLL + YLD**.
5. Web of causation – **chronic diseases**.
6. Immunization – **primary prevention**.
7. Screening – **secondary prevention**.
8. Rehabilitation – **tertiary prevention**.
9. Prevent risk factors – **primordial**.
10. Iceberg phenomenon – **hypertension, diabetes**.
11. Only eradicated disease – **smallpox**.
12. Health Survey & Development Committee – **Bhore**.
13. Multipurpose workers – **Kartar Singh**.
14. Village health guides – **Shrivastava**.
15. Integration – **Jungalwalla**.
16. Alma-Ata goal – **Health for All by 2000**.
17. PHC principles – **equity, participation, intersectoral, appropriate technology**.
18. Health is – **State subject**.
19. SC plain – **5,000**.
20. PHC plain – **30,000**.
21. CHC – **1,20,000, 30 beds**.
22. FRU – **CHC**.
23. ASHA – **1 per 1,000**.
24. HWC renamed – **Ayushman Arogya Mandir**.
25. CPHC packages – **12**.
26. PMJAY – **₹5 lakh**.
27. NRHM – **12 April 2005**.
28. Father of modern epidemiology – **John Snow**.
29. Constant presence – **endemic**.
30. Animal to man – **zoonosis**.
31. Ratio – **numerator not in denominator**.
32. P = **I × D**.
33. Killing power – **CFR**.
34. Infectiousness – **SAR**.
35. Odds ratio – **case-control**.
36. Relative risk – **cohort**.
37. Gold standard study – **RCT**.
38. Sensitivity – **TP/(TP+FN)**.
39. Specificity – **TN/(TN+FP)**.
40. First step epidemic investigation – **verify diagnosis**.

41. Best dispersion – **SD**.
42. Trend – **line diagram**.
43. Continuous data – **histogram**.
44. Correlation – **scatter diagram**.
45. Immunization coverage survey – **30-cluster**.
46. Qualitative data test – **chi-square**.
47. Father of demography – **John Graunt**.
48. India demographic stage – **late expanding**.
49. Census interval – **10 years**.
50. Birth registration – **21 days**.

One-liners 51–100

51. SRS – **dual record system**.
52. Replacement TFR – **2.1**.
53. Census 2011 sex ratio – **943**.
54. Carrier without disease – **healthy carrier**.
55. Chronic carrier – **typhoid**.
56. Quarantine period – **longest IP**.
57. Isolation – **infected persons**.
58. Malaria in mosquito – **cyclo-propagative**.
59. Hand washing – **most effective infection control**.
60. Concurrent disinfection – **during illness**.
61. Terminal disinfection – **after patient leaves**.
62. Phenol – **Rideal-Walker coefficient**.
63. Blood spill – **1 % hypochlorite**.
64. Endoscopes – **2 % glutaraldehyde**.
65. IDSP forms – **S, P, L**.
66. World TB Day – **24 March**.
67. TB diagnosis – **NAAT**.
68. INH side effect – **neuropathy**.
69. Ethambutol – **optic neuritis**.
70. Leprosy nerve – **ulnar**.
71. Clofazimine – **skin pigmentation**.
72. Anti-Leprosy Day – **30 Jan**.
73. Dengue vector – **Aedes**.
74. Filaria vector – **Culex**.
75. Kala-azar vector – **sandfly**.
76. JE amplifier – **pigs**.
77. Larvivorous fish – **Gambusia**.
78. Temephos – **larvicide (drinking water)**.
79. HIV targets – **95-95-95**.
80. World AIDS Day – **1 Dec**.
81. Rabies category III – **vaccine + RIG**.
82. NCD screening age – **≥ 30**.
83. VIA – **acetic acid**.
84. Blindness cause – **cataract**.
85. Tele-MANAS – **14416**.
86. COTPA public smoking ban – **Section 4**.
87. Tobacco sale near school – **100 yards**.
88. Elderly – **≥ 60 yrs**.
89. Iodised salt (consumer) – **15 ppm**.
90. Fluorosis – **> 1.5 mg/L**.
91. Defluoridation – **Nalgonda technique**.
92. RPwD – **21 disabilities**.

93. World Health Day – **7 April**.
94. Nurses Day – **12 May**.
95. India FP programme – **1952**.
96. Most popular FP method – **female sterilisation**.
97. MTP ≤ 20 wks – **1 RMP**.
98. PCPNDT – **1994**.
99. Maternity leave – **26 weeks**.
100. ICDS services – **6**.

One-liners 101–150

101. ICDS child supplementary – **500 kcal, 12–15 g protein**.
102. PM POSHAN primary – **450 kcal, 12 g protein**.
103. Reference protein – **egg**.
104. Asian obesity BMI – **≥ 25**.
105. Lathyrism – **Kesari dal**.
106. Epidemic dropsy – **argemone oil**.
107. Aflatoxin – **liver cancer**.
108. Pasteurisation test – **phosphatase**.
109. FSSAI Act – **2006**.
110. Vital layer – **slow sand filter**.
111. Backwashing – **rapid sand filter**.
112. Horrock's – **chlorine demand**.
113. OT test – **residual chlorine**.
114. Best household purification – **boiling**.
115. Faecal indicator – **E. coli**.
116. Blue baby – **nitrates**.
117. Temporary hardness – **boiling**.
118. Water seal – **2 cm**.
119. ODF India – **2019**.
120. BOD – **organic pollution**.
121. City refuse – **sanitary landfill**.
122. Anatomical waste – **yellow**.
123. IV sets/ gloves – **red**.
124. Needles – **white translucent**.
125. Broken glass – **blue**.
126. Plague – **rat flea**.
127. KFD – **tick**.
128. Scrub typhus – **mite**.
129. Silicosis – **TB risk**.
130. Byssinosis – **cotton**.
131. Farmer's lung – **mouldy hay**.
132. Lead – **Burtonian line**.
133. Aniline dyes – **bladder cancer**.
134. ESI Act – **1948**.
135. Factories Act crèche – **≥ 30 women**.
136. NIOH – **Ahmedabad**.
137. Behaviour change – **individual approach**.
138. Group discussion – **6–12**.
139. Panel – **experts discuss before audience**.
140. Symposium – **series of speeches**.
141. Most concrete experience – **direct purposeful**.
142. FP counselling – **GATHER**.

143. Backbone of CHN – **home visit**.
144. Bag technique – **prevent cross-infection**.
145. First school health – **Baroda 1909**.
146. DM Act – **2005**; NDMA – **PM**.
147. Red tag – **immediate**.
148. 108 in Gujarat – **2007**.
149. WHO SEARO – **New Delhi**.
150. GOBI-FFF – **UNICEF**.



Exam strategy – Asha Didi ki last advice

- GSSSB paper me **Community Health se ~10–15 questions** aate hain – **health care delivery norms, committees, national programmes, epidemiology terms, water/ BMW, health education methods aur Gujarat schemes** sabse important hain.
- **Years aur numbers** wali tables (17.1, 17.2) roz 10 minute revise karo – ye seedhe MCQ banti hain.
- National programmes me **launch year + target + main drug/ test** – teeno yaad rakho.
- MCQ Bank PDF ke saare questions **2 baar** solve karo, galtiyan notebook me likho.

— Community Health Nursing · GSSSB Staff Nurse Preparation · Padhte raho, revise karte raho! —